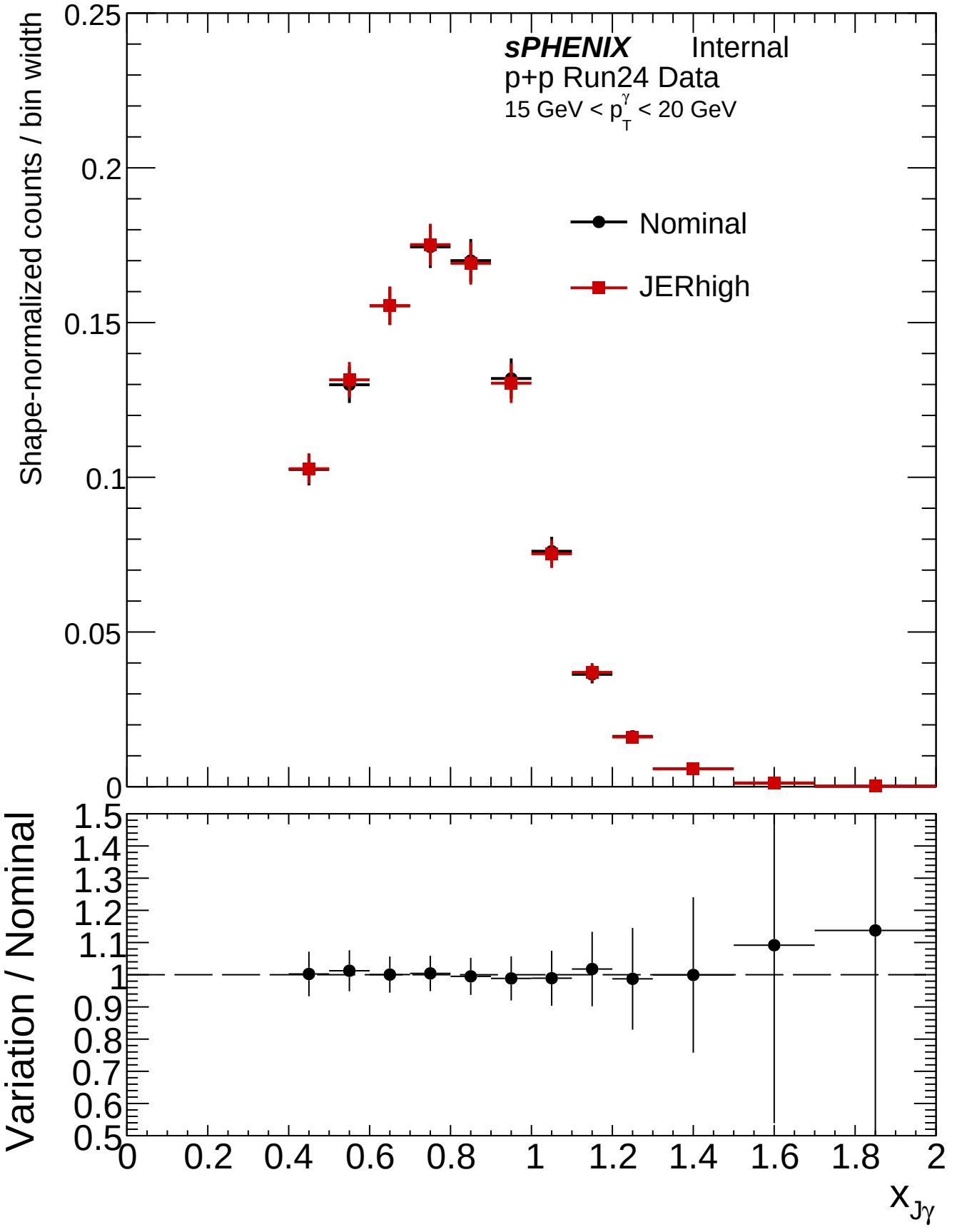
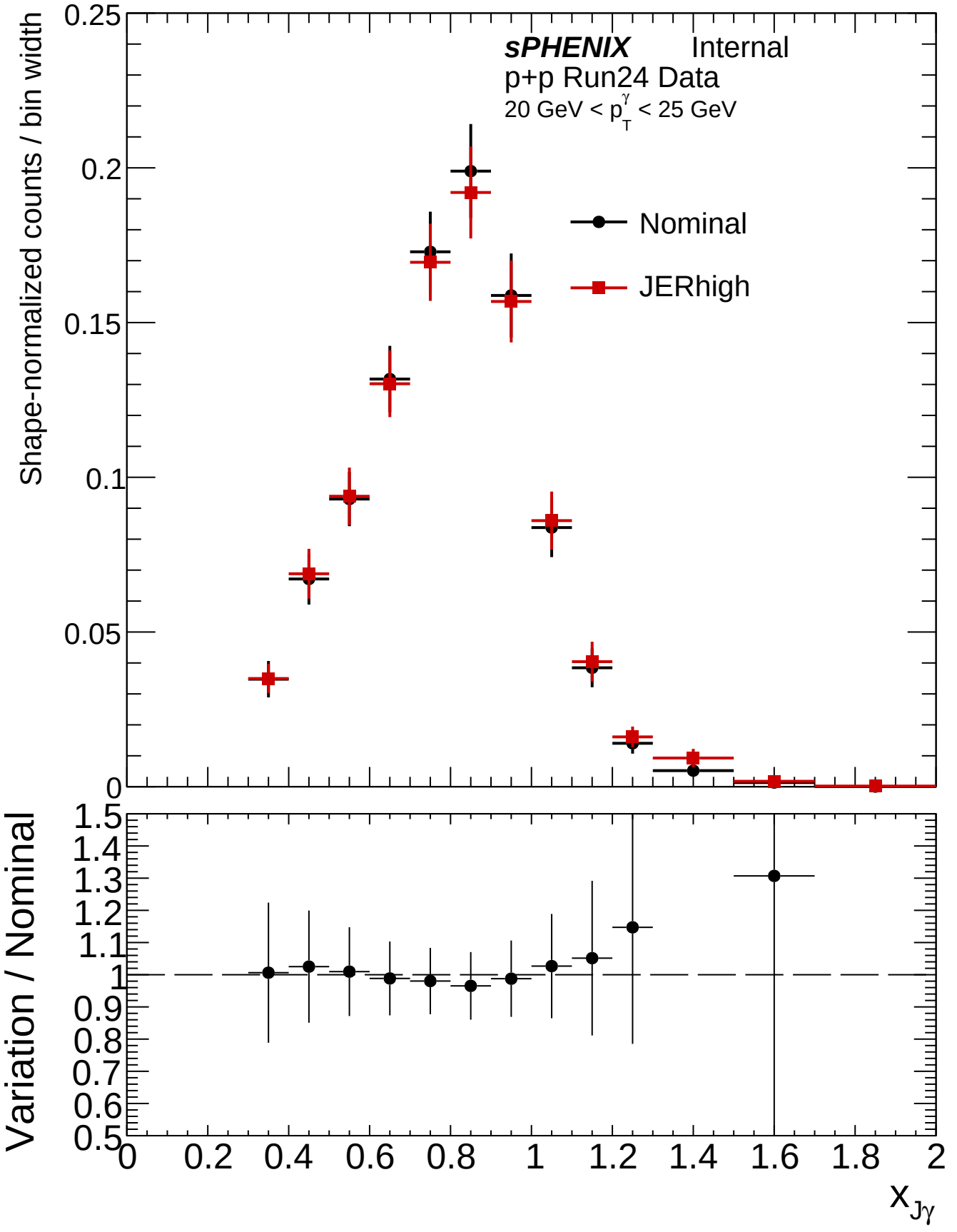

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
JERhigh
Shape-normalized counts / bin width
0.25
0.2
0.15
0.1
0.05
0
Variation / Nominal
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
x_Jγ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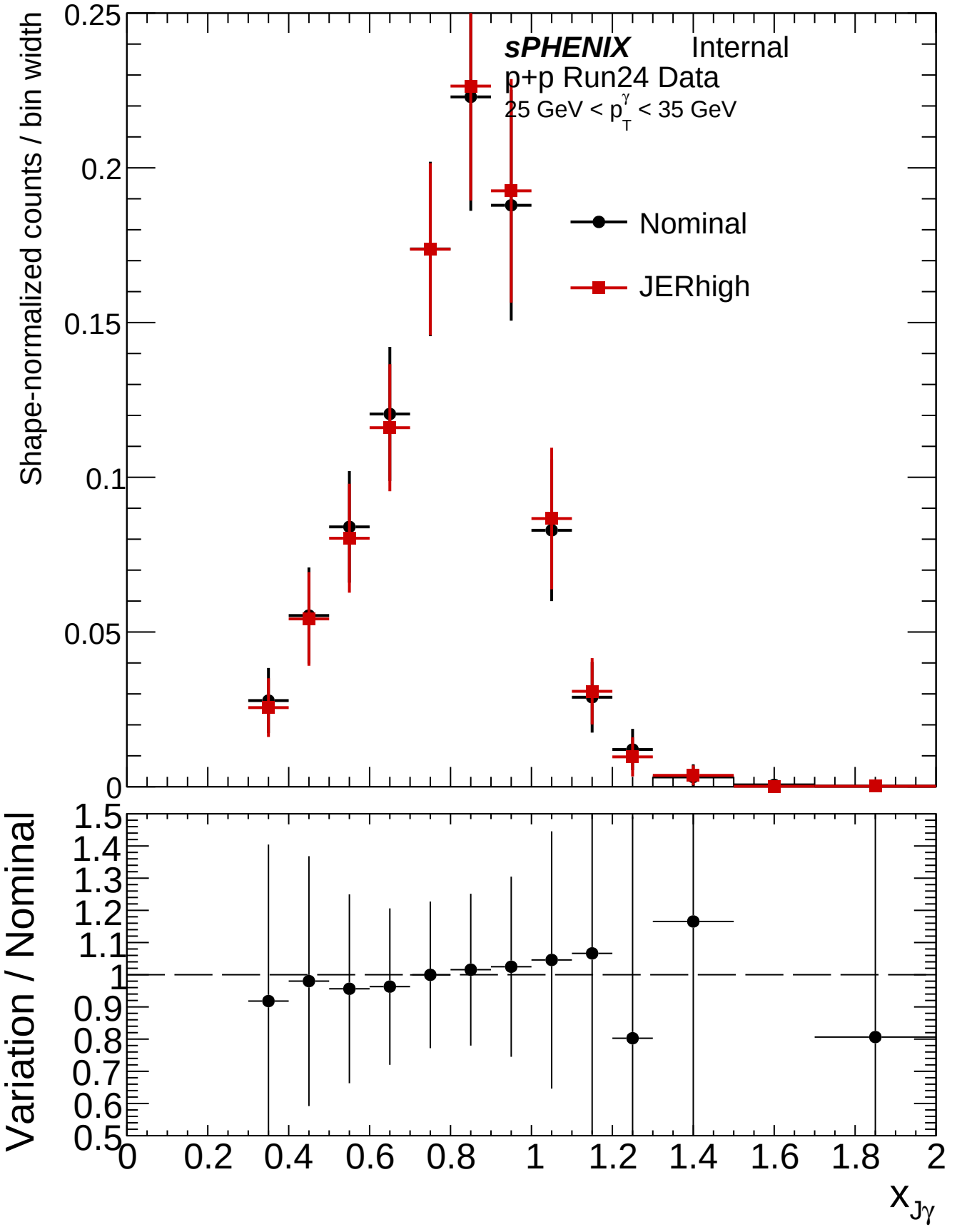
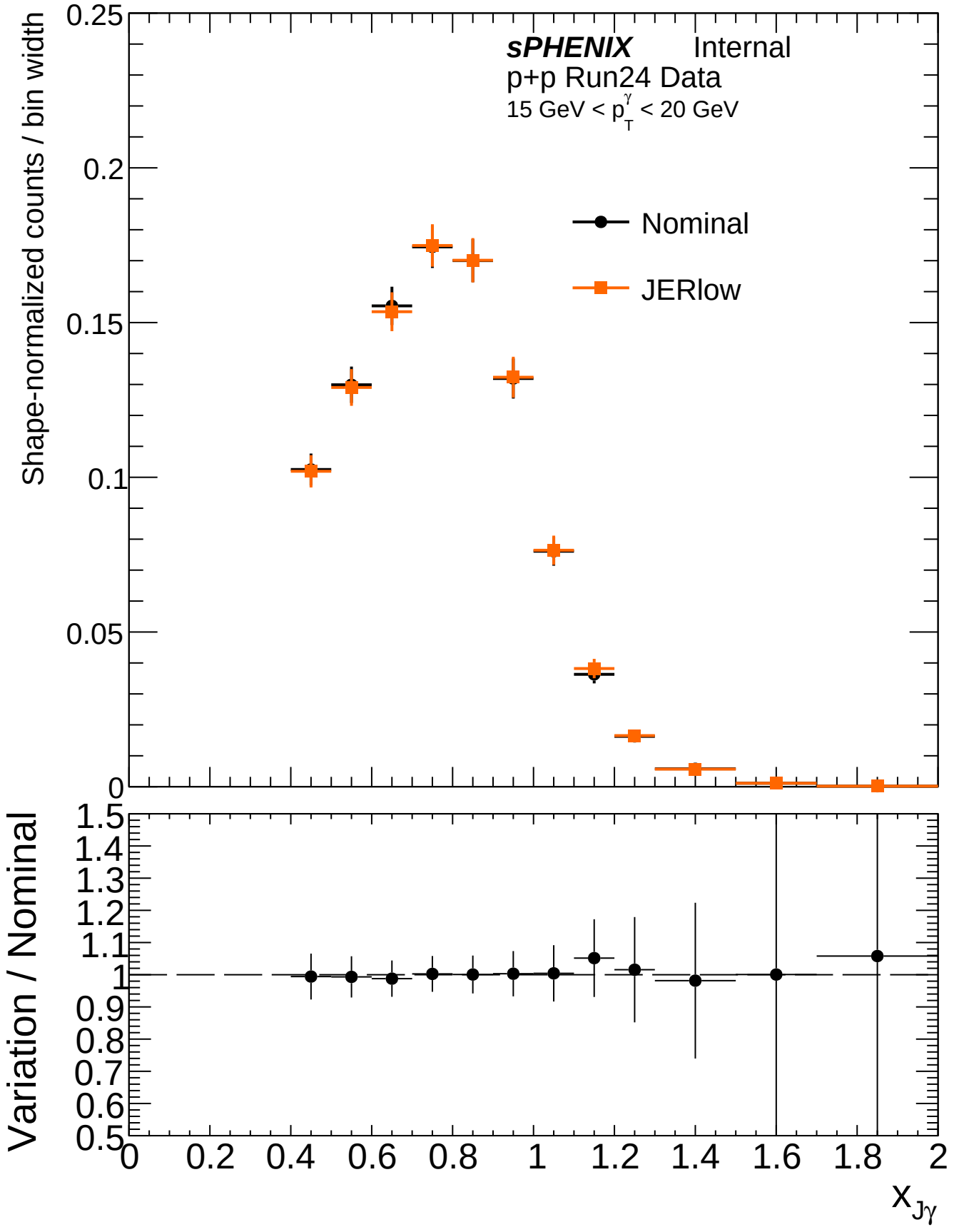
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
JERhigh
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
JERhigh
Shape-normalized counts / bin width
Variation / Nominal
$x_{J\gamma}$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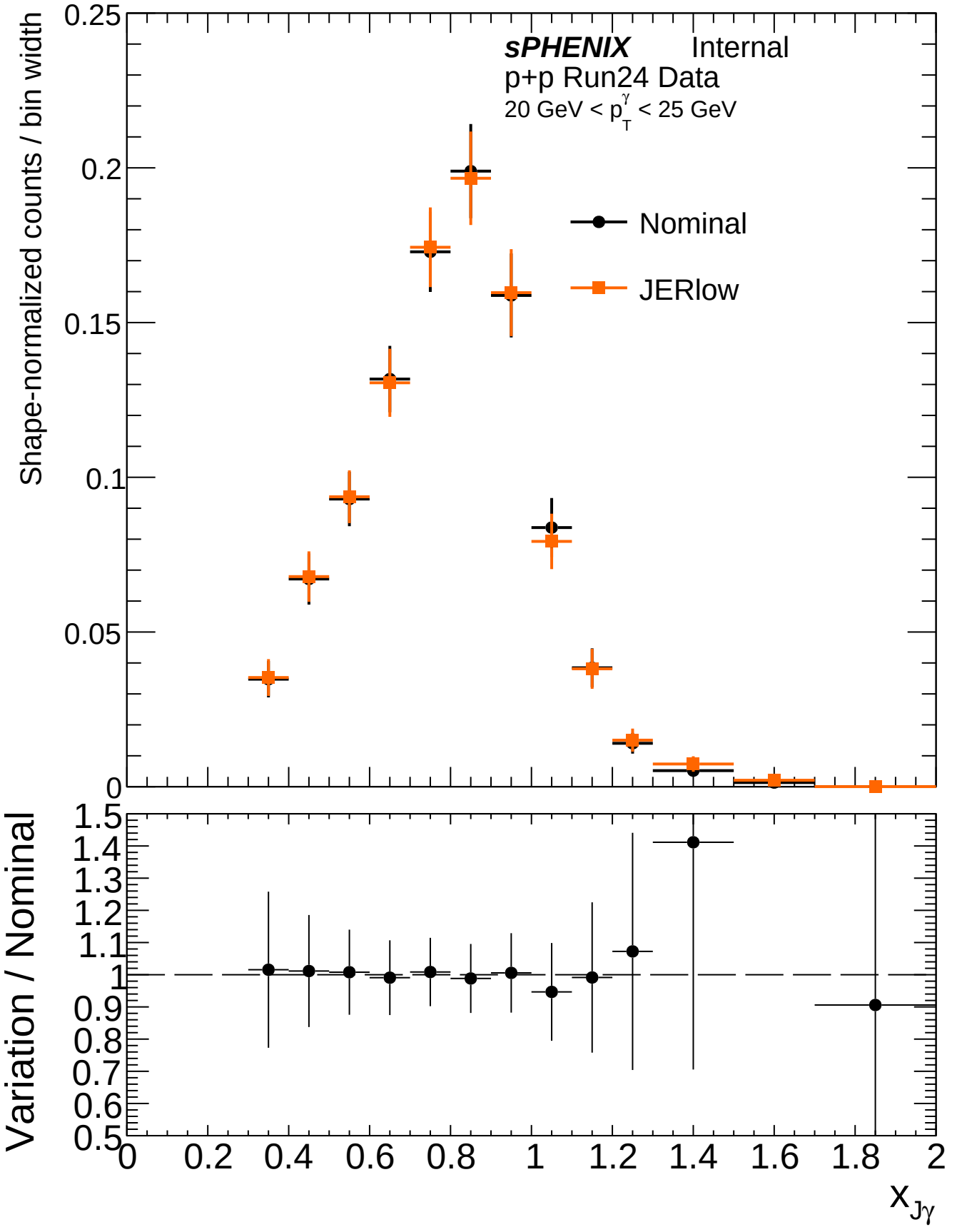
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
JERlow
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ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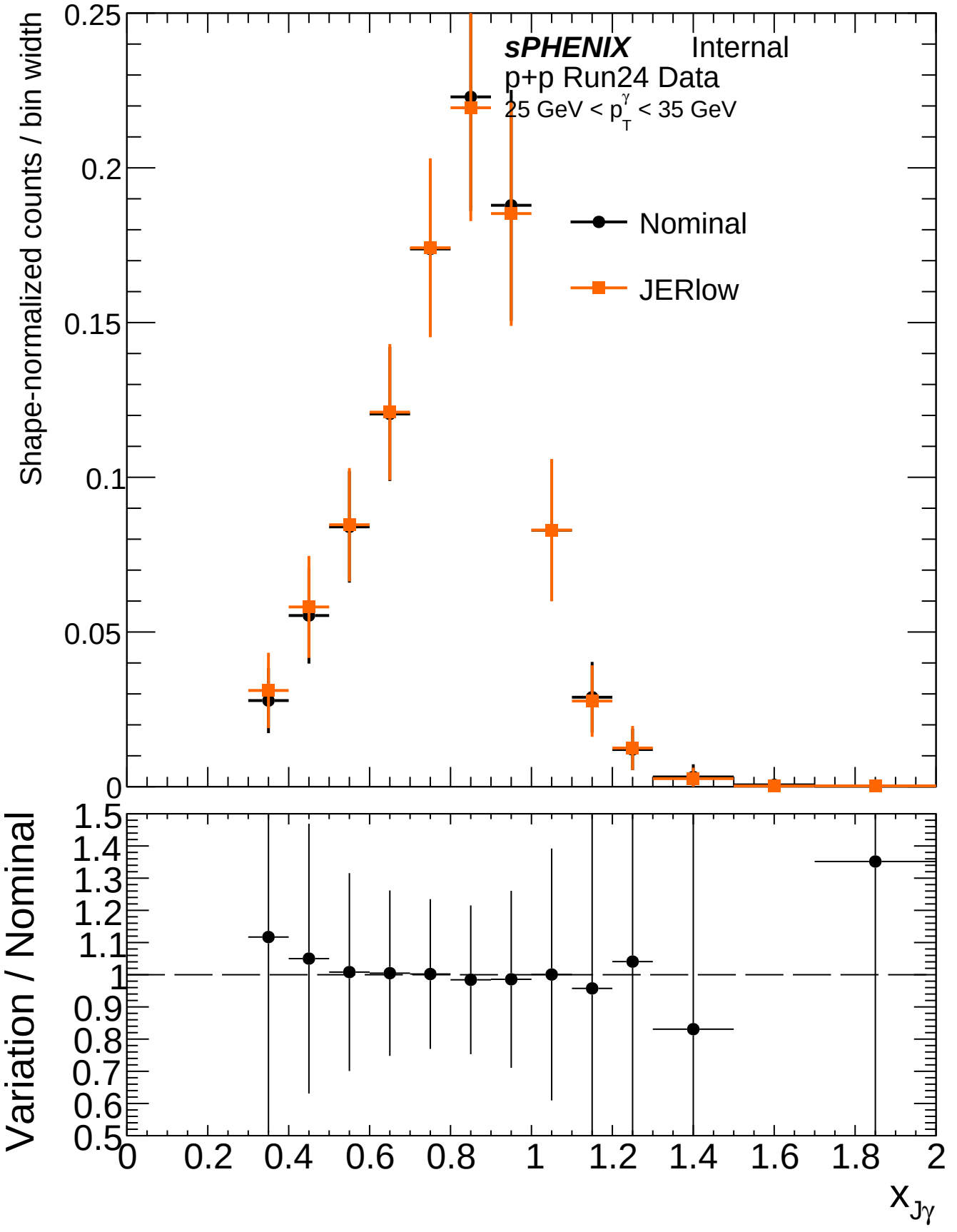
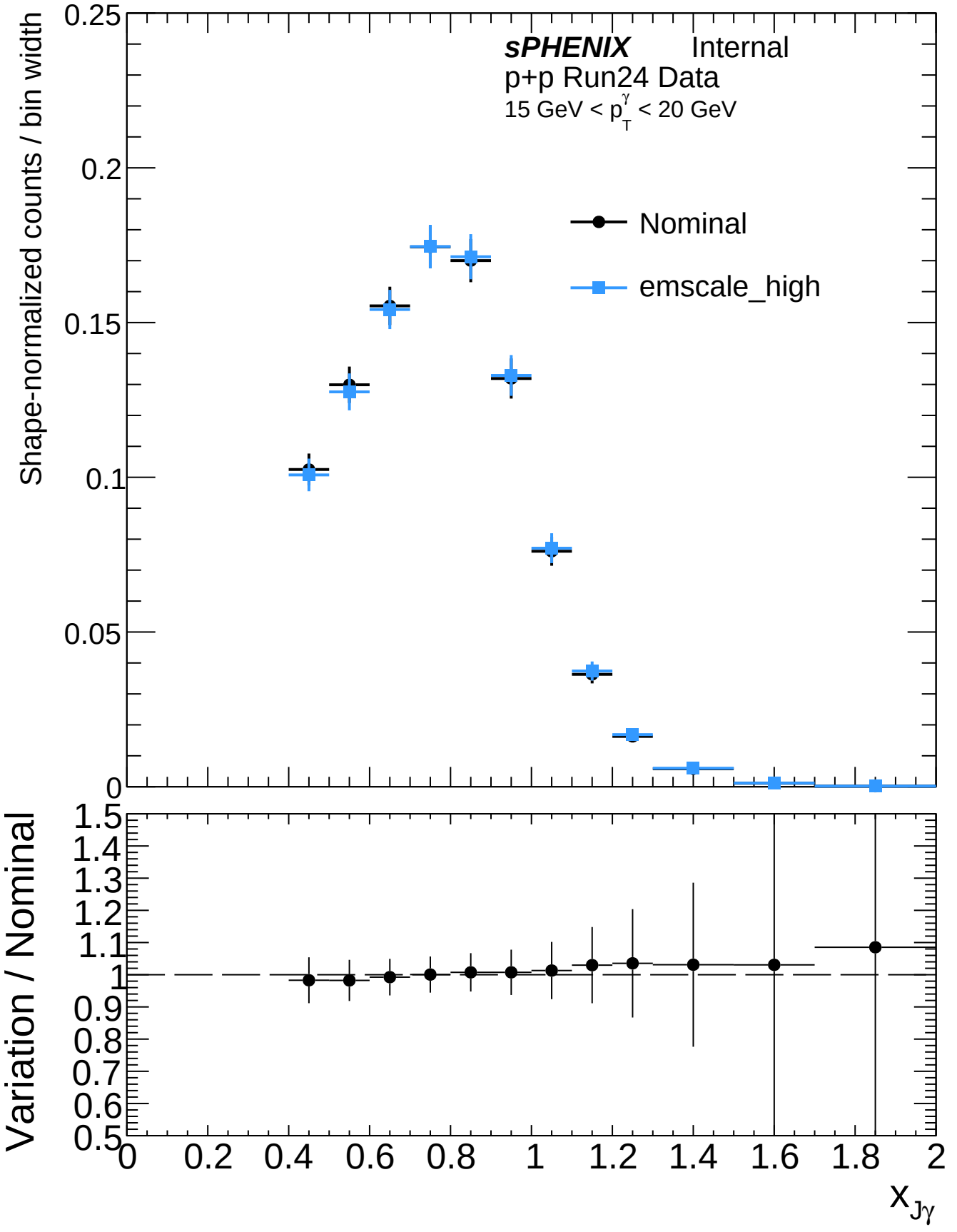
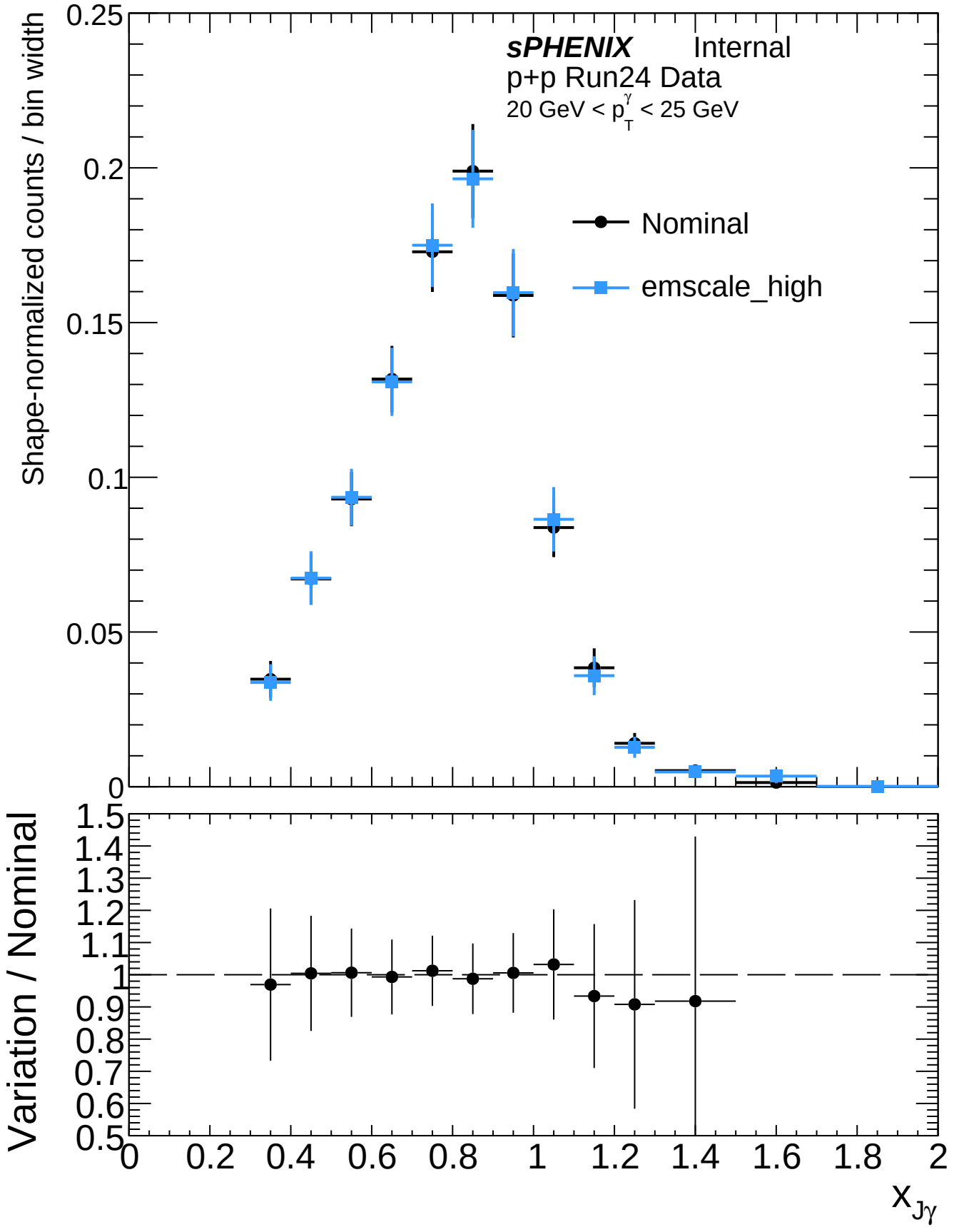
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
JERlow
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ


sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
JERlow
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
emscale_high
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ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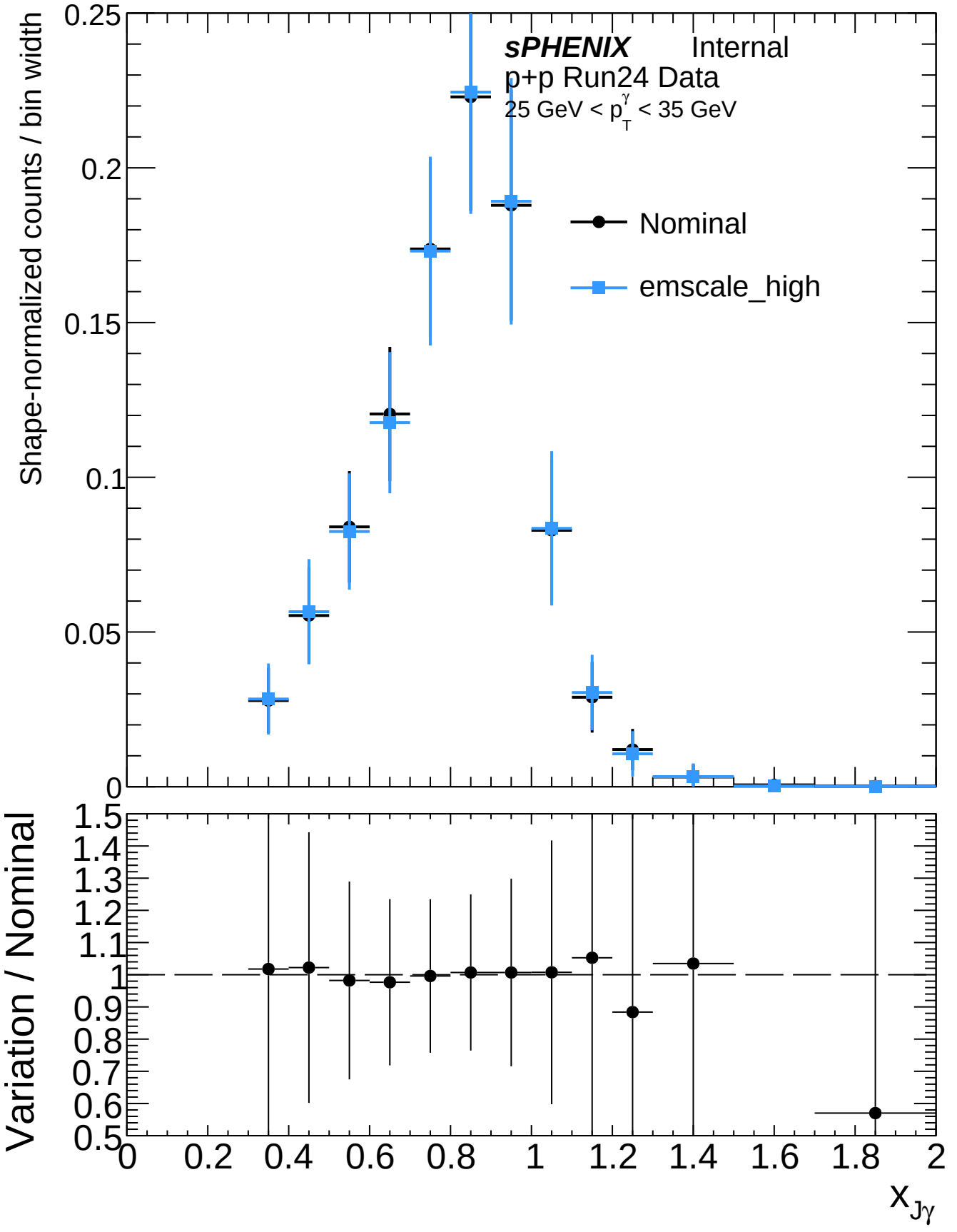
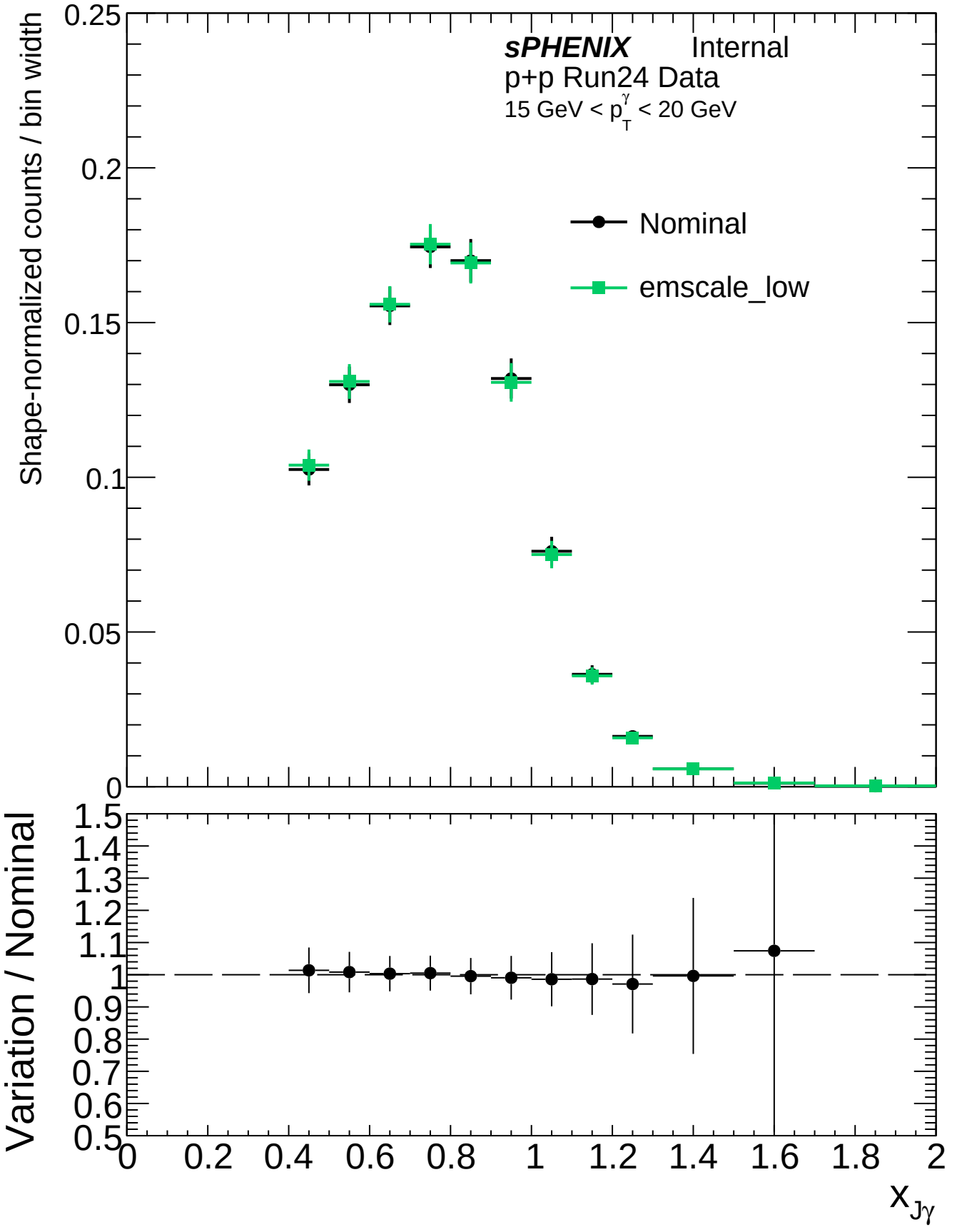
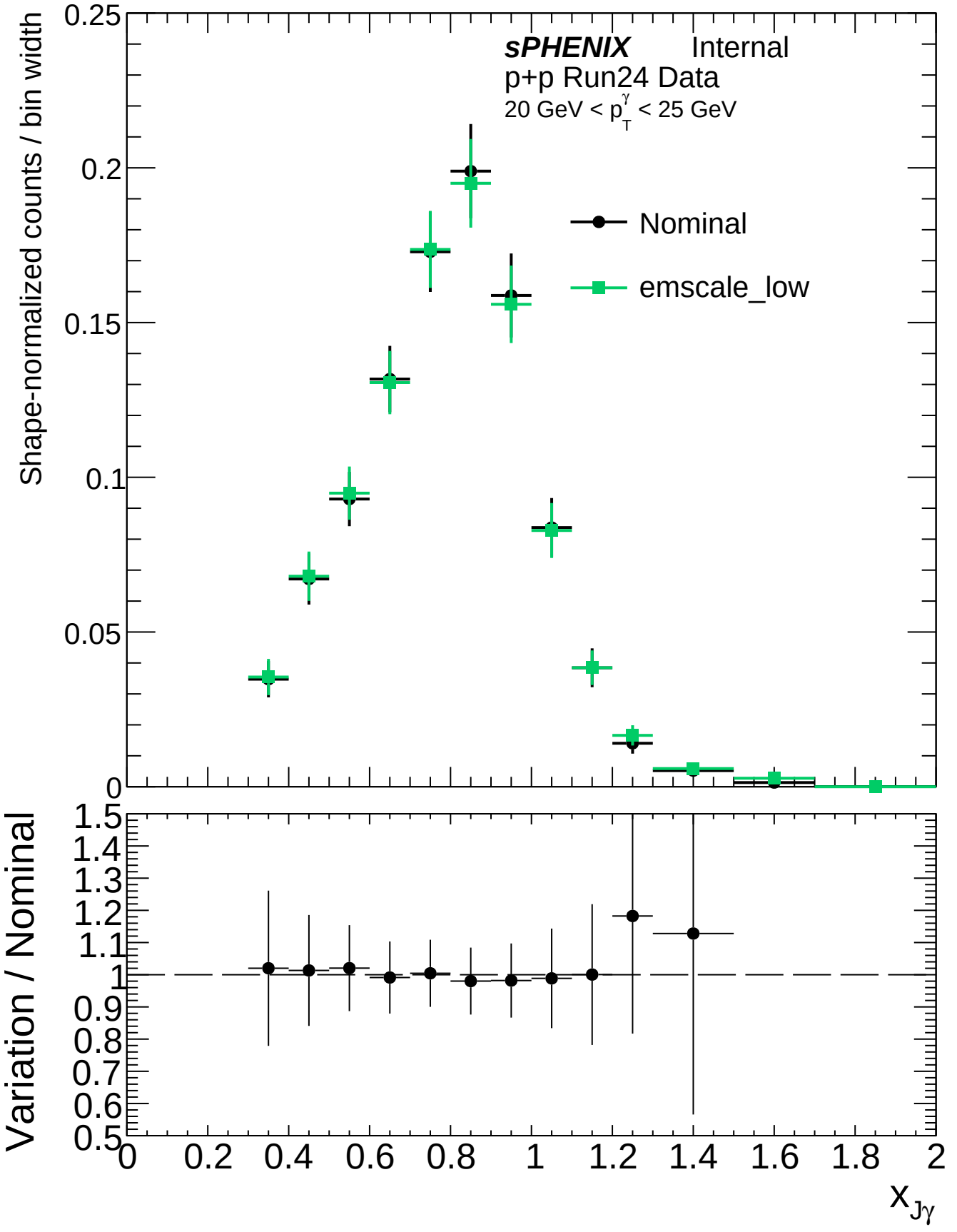
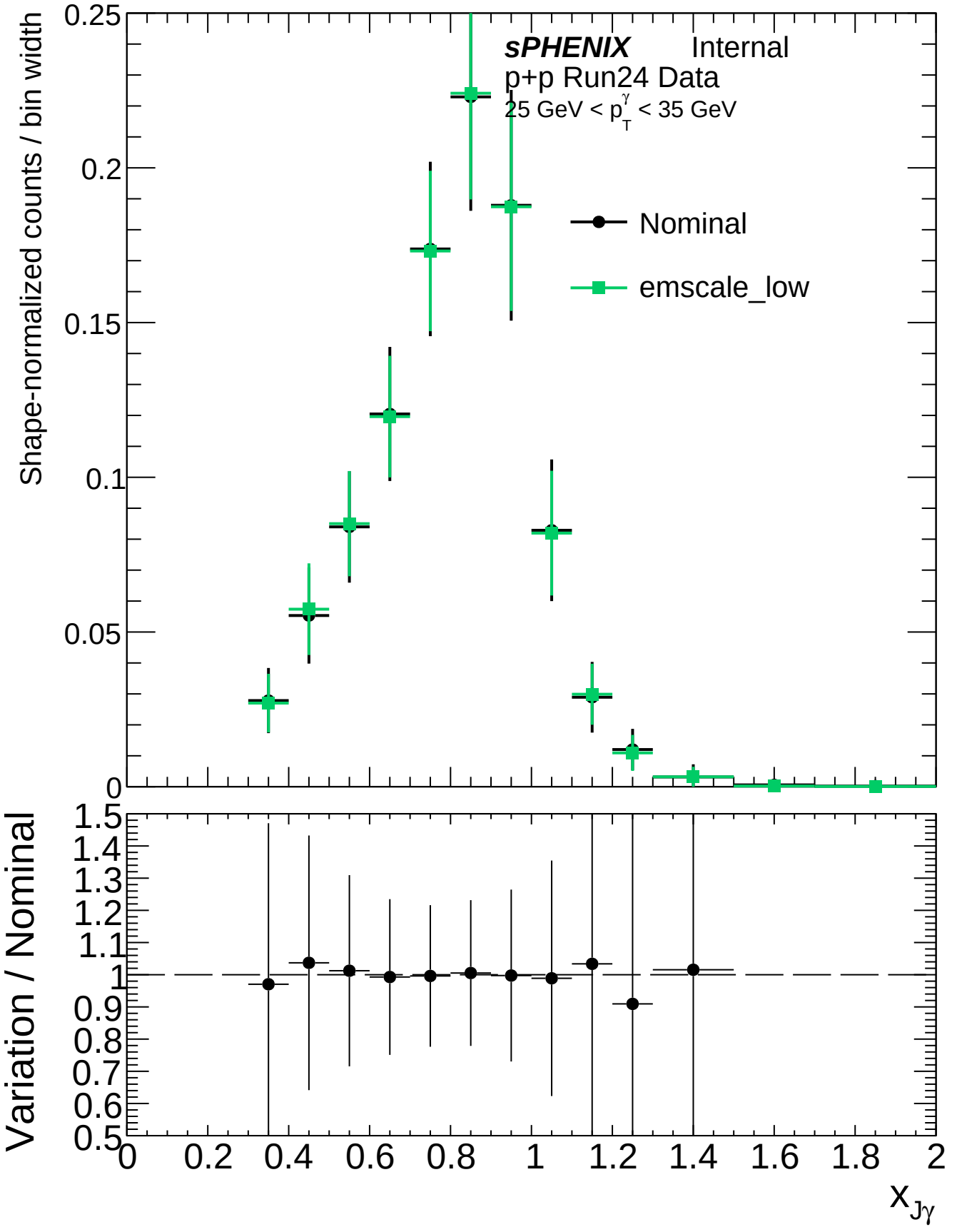
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
emscale_high
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ


sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
emscale_high
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
emscale_low
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ


sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
emscale_low
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
emscale_low
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
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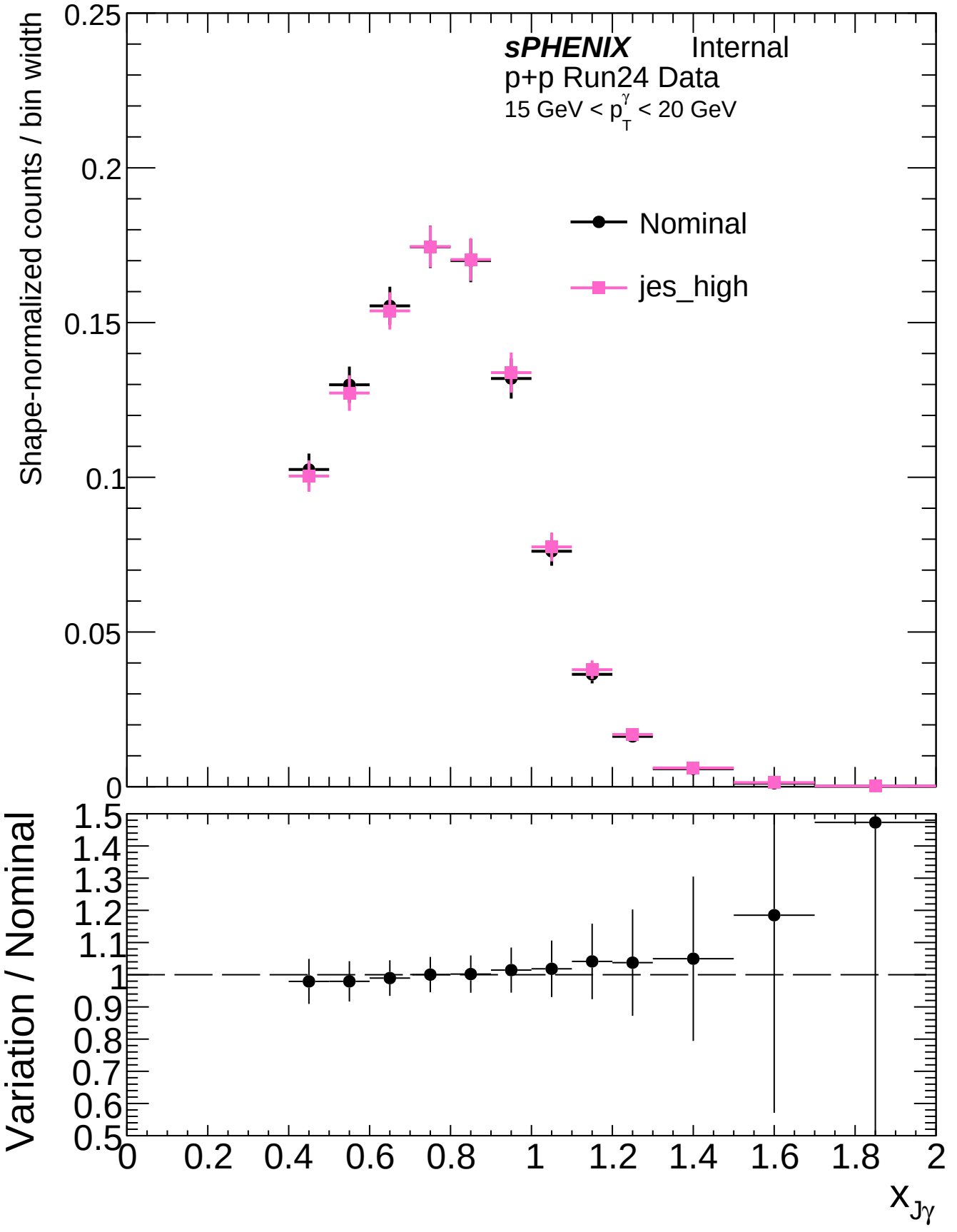
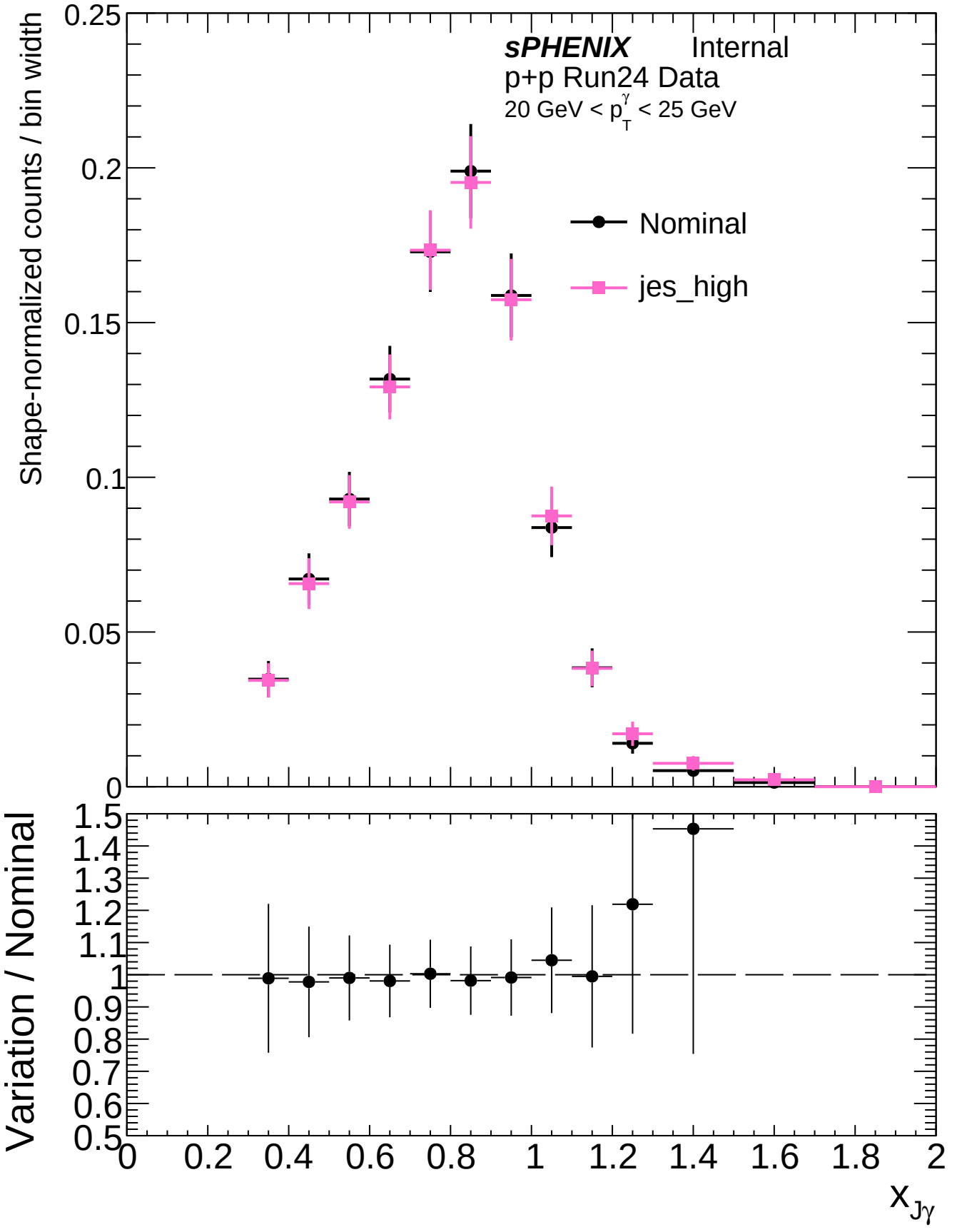
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
jes_high
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ


sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
jes_high
Shape-normalized counts / bin width
Variation / Nominal
$x_{J\gamma}$
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
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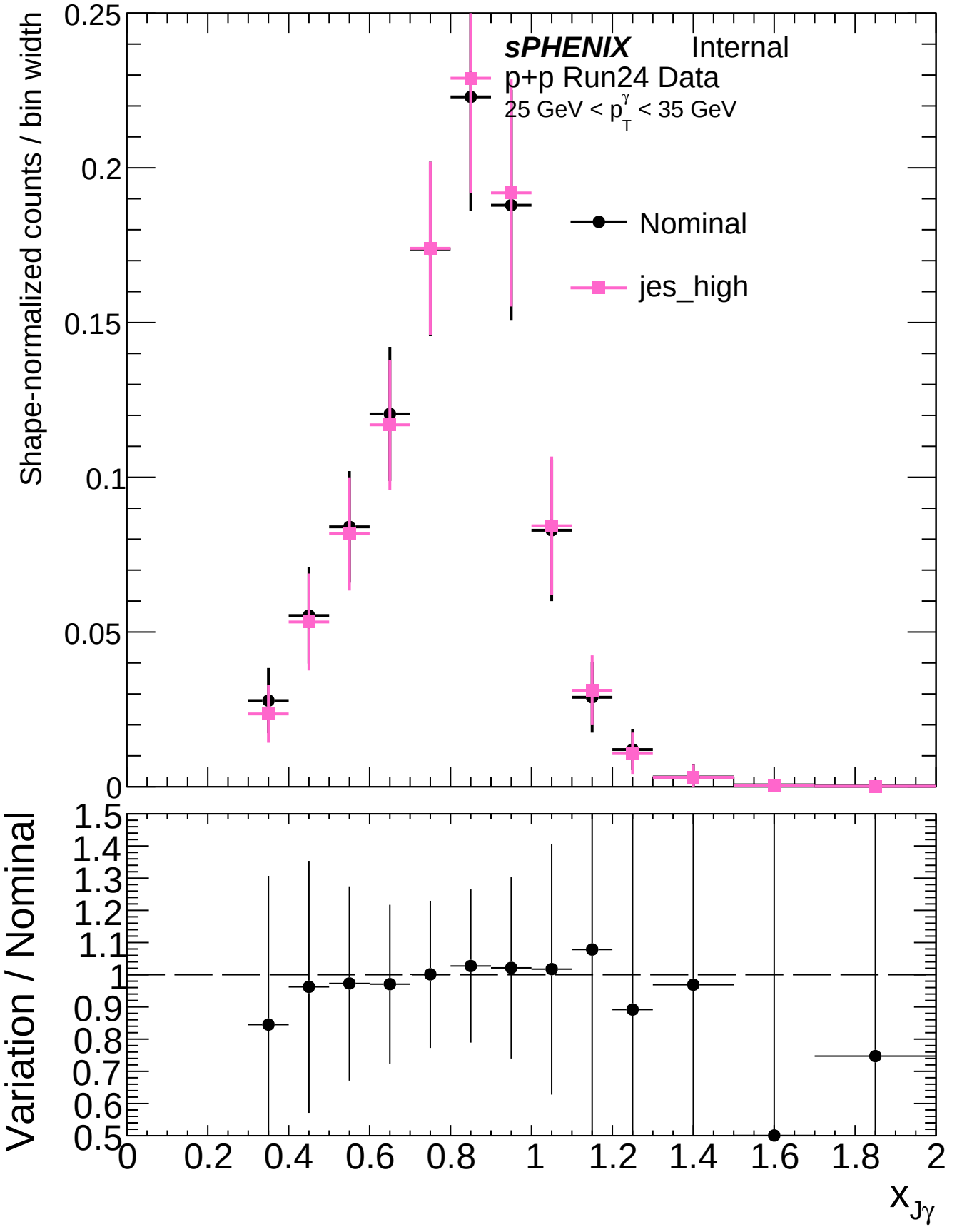
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
jes_high
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
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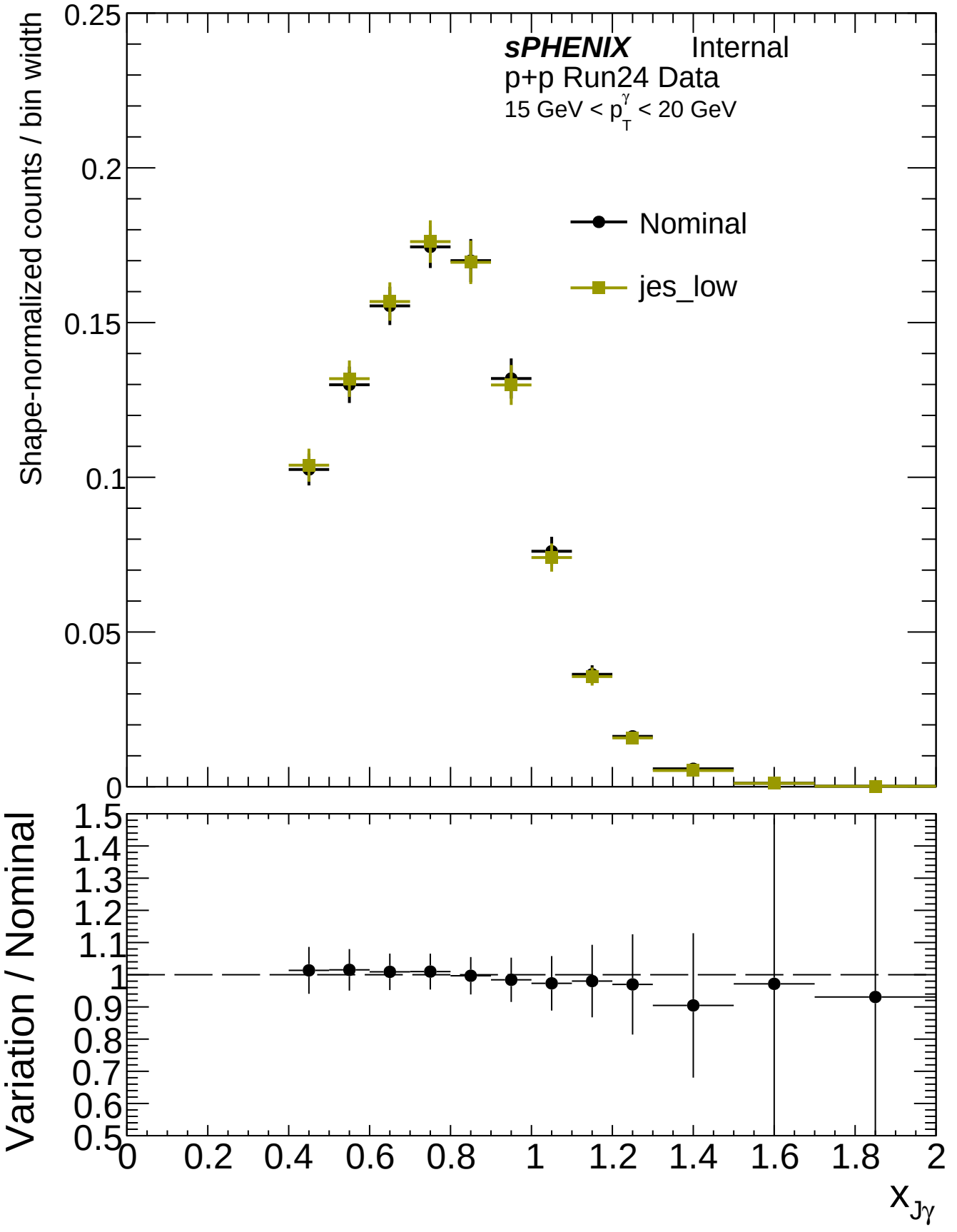
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
jes_low
Shape-normalized counts / bin width
0.25
0.2
0.15
0.1
0.05
0
Variation / Nominal
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
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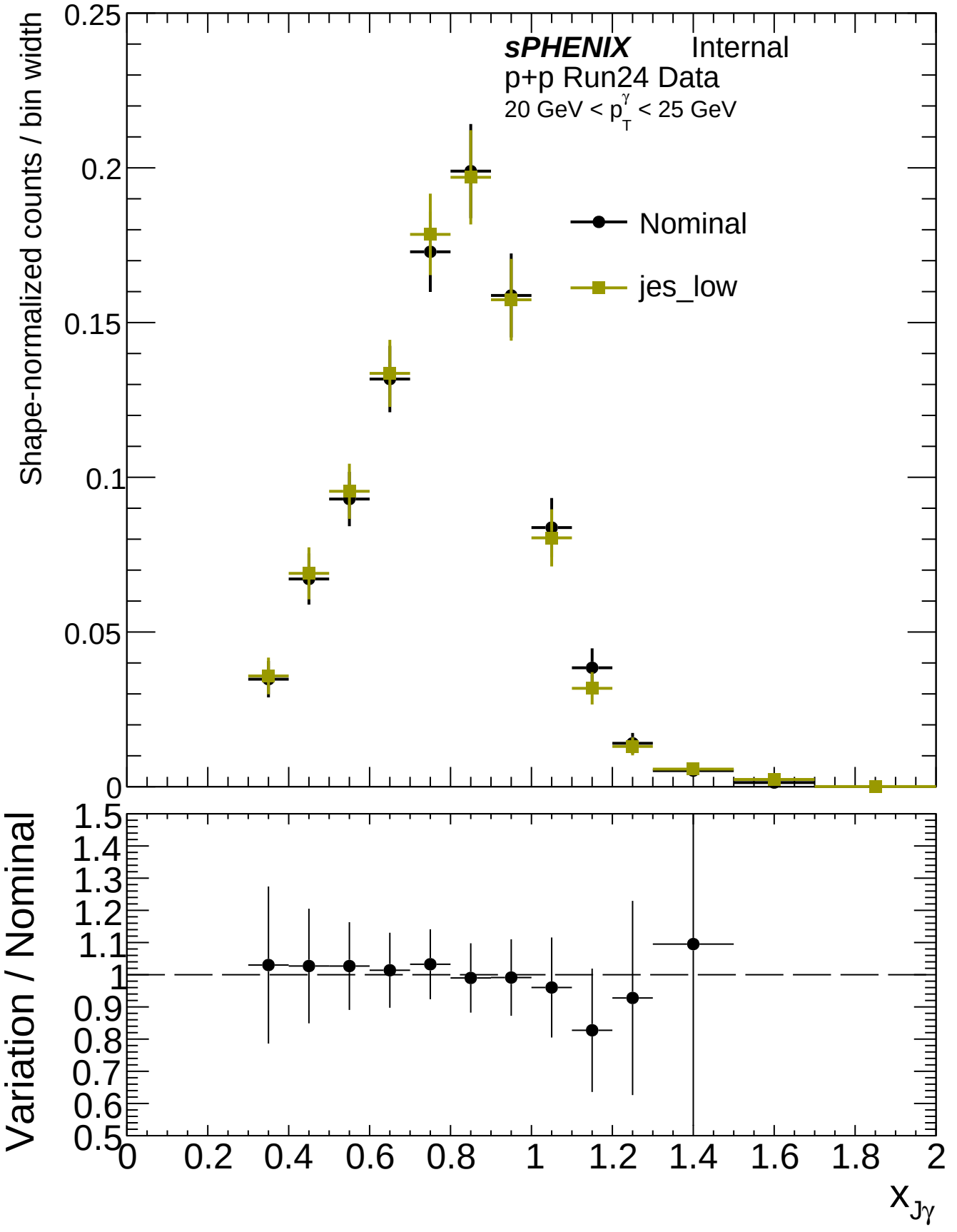
0.8
1
1.2
1.4
1.6
1.8
2
x_Jγ


sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
jes_low
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
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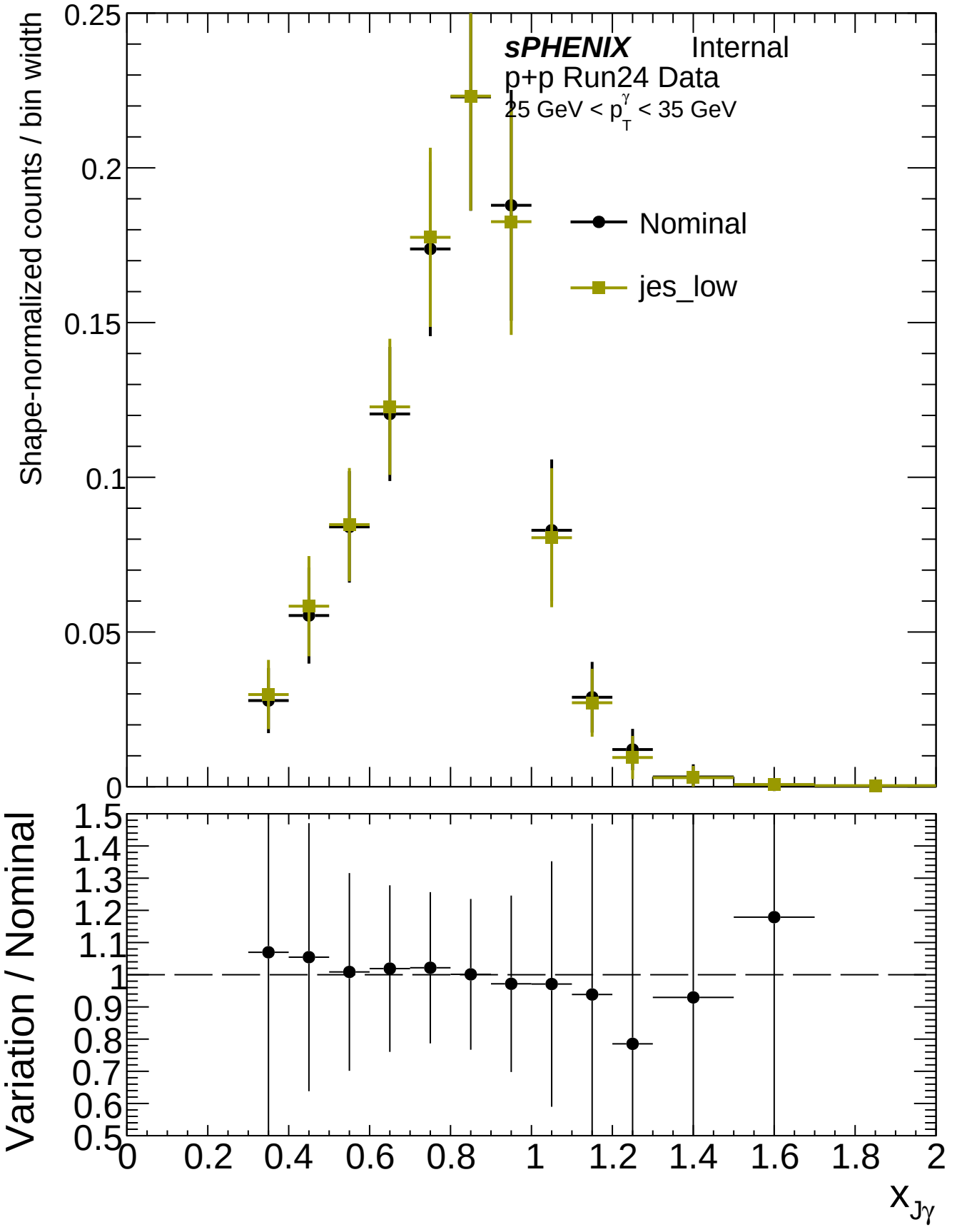
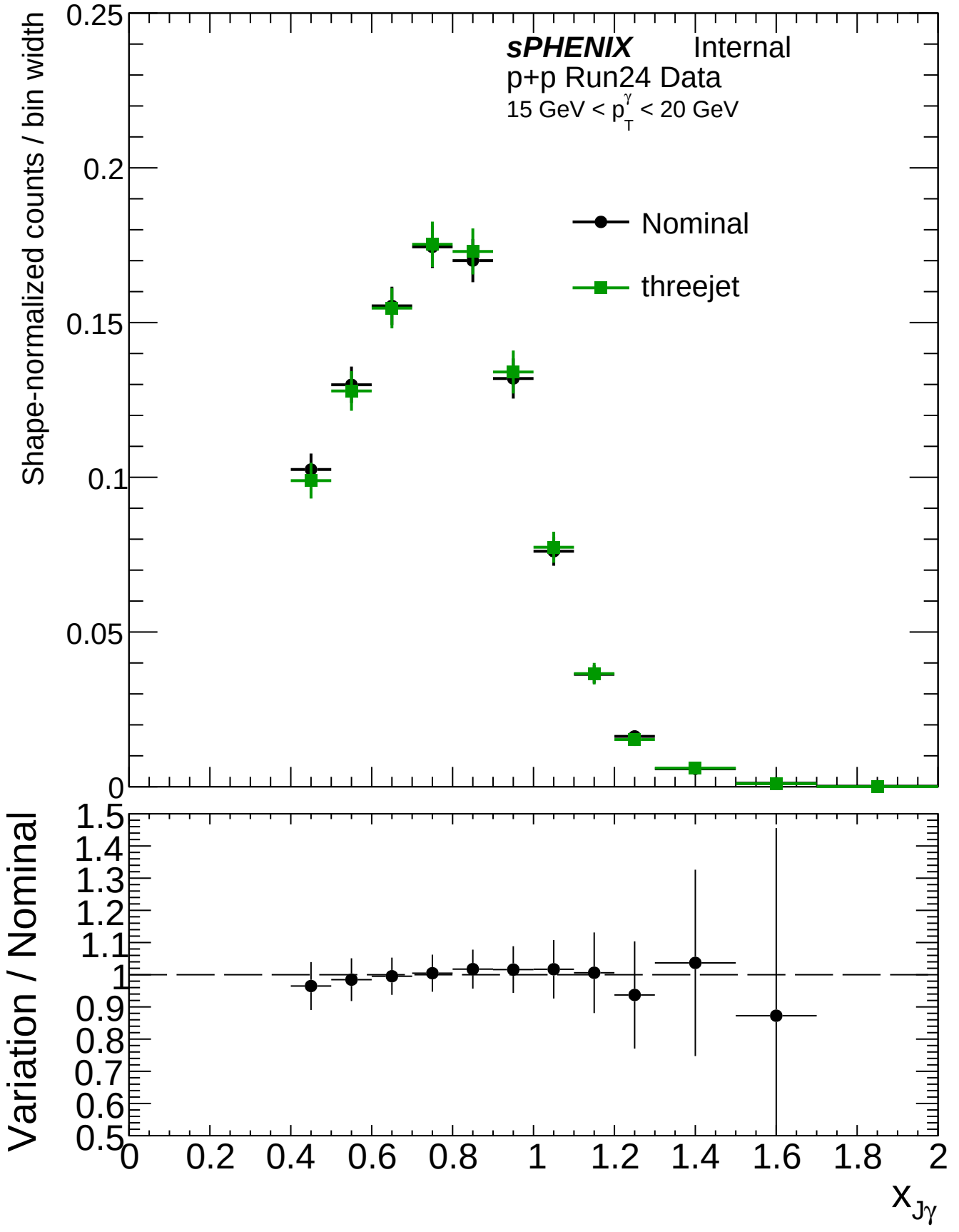
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
jes_low
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ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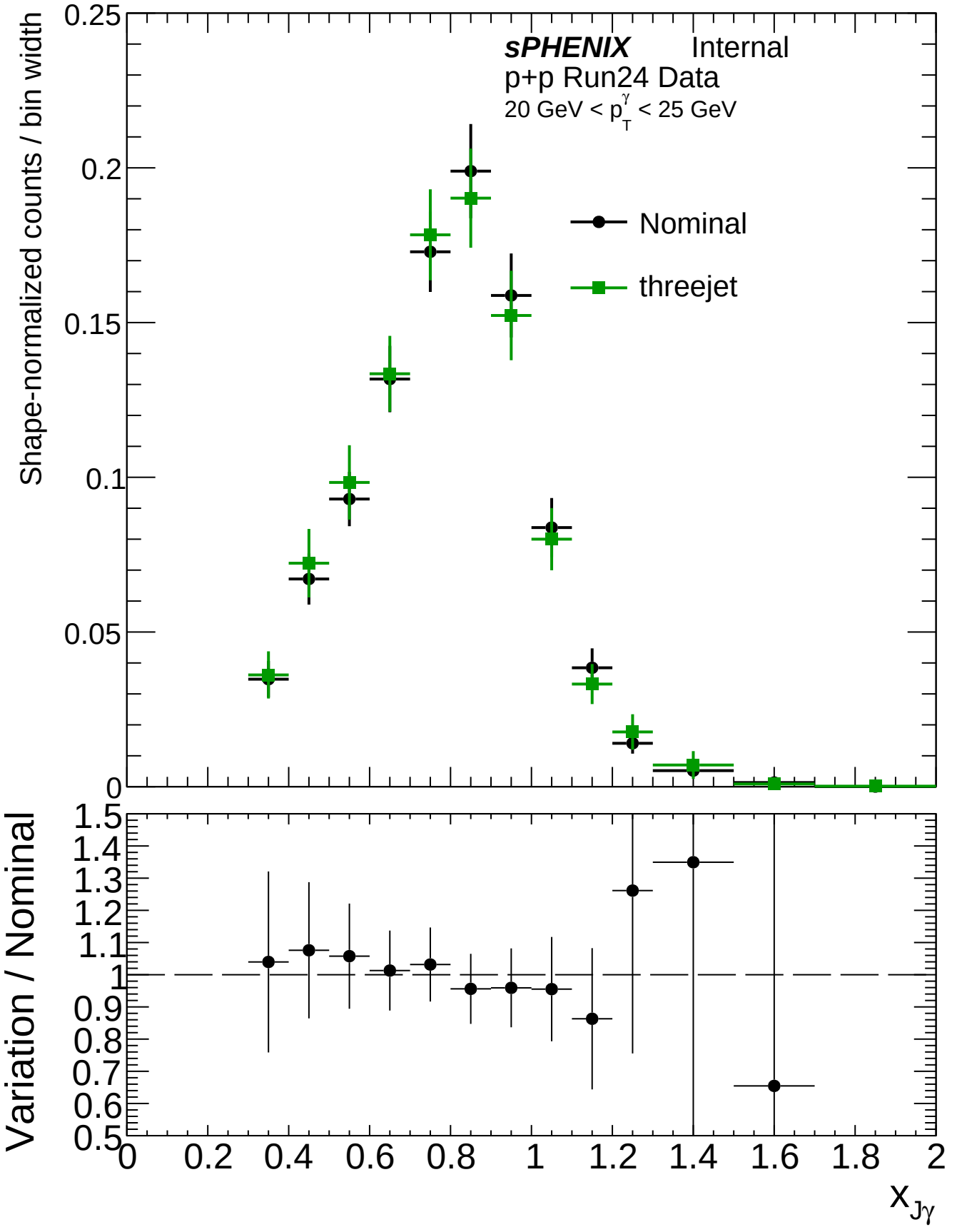
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
threejet
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ


sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
threejet
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
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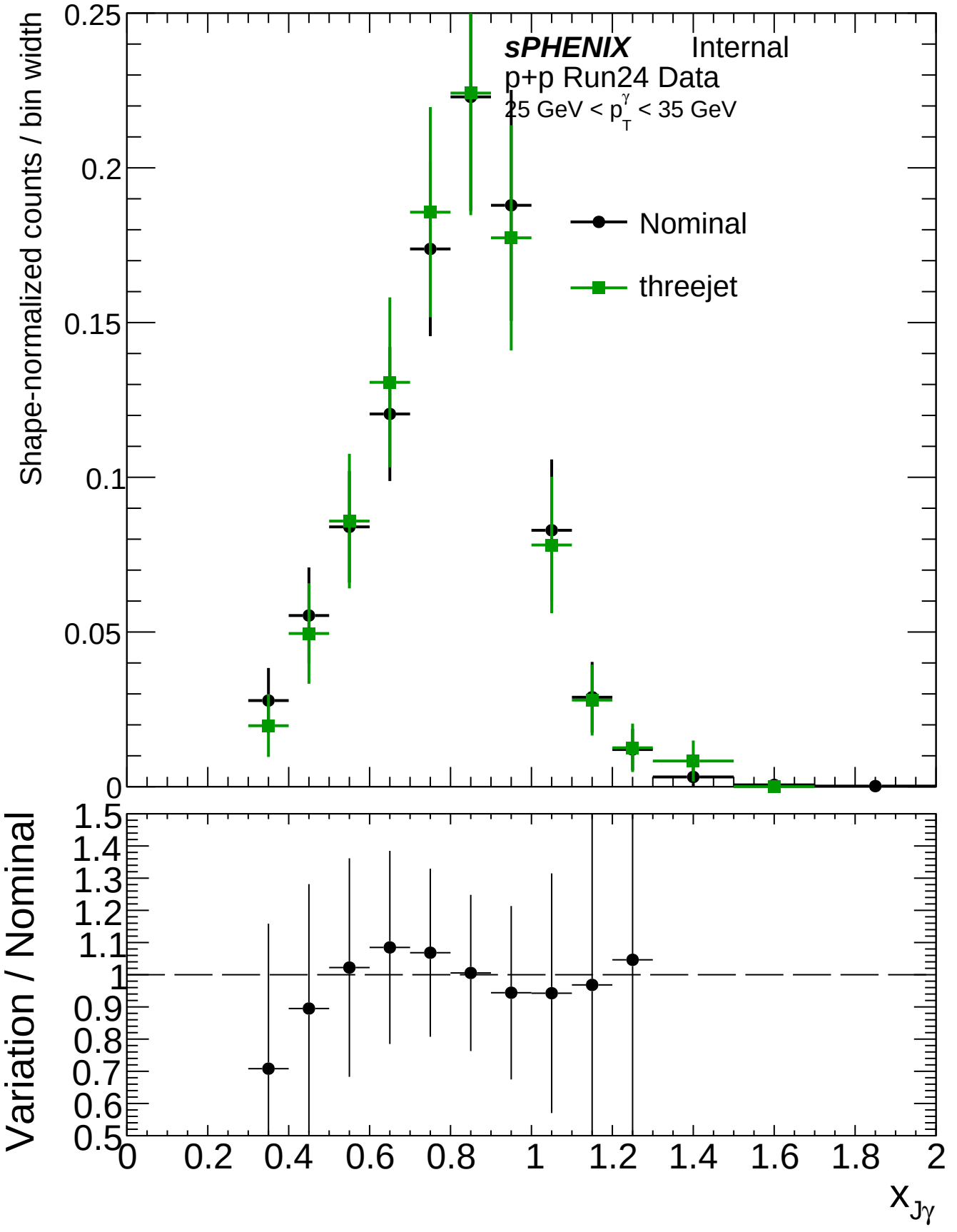
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
threejet
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
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0
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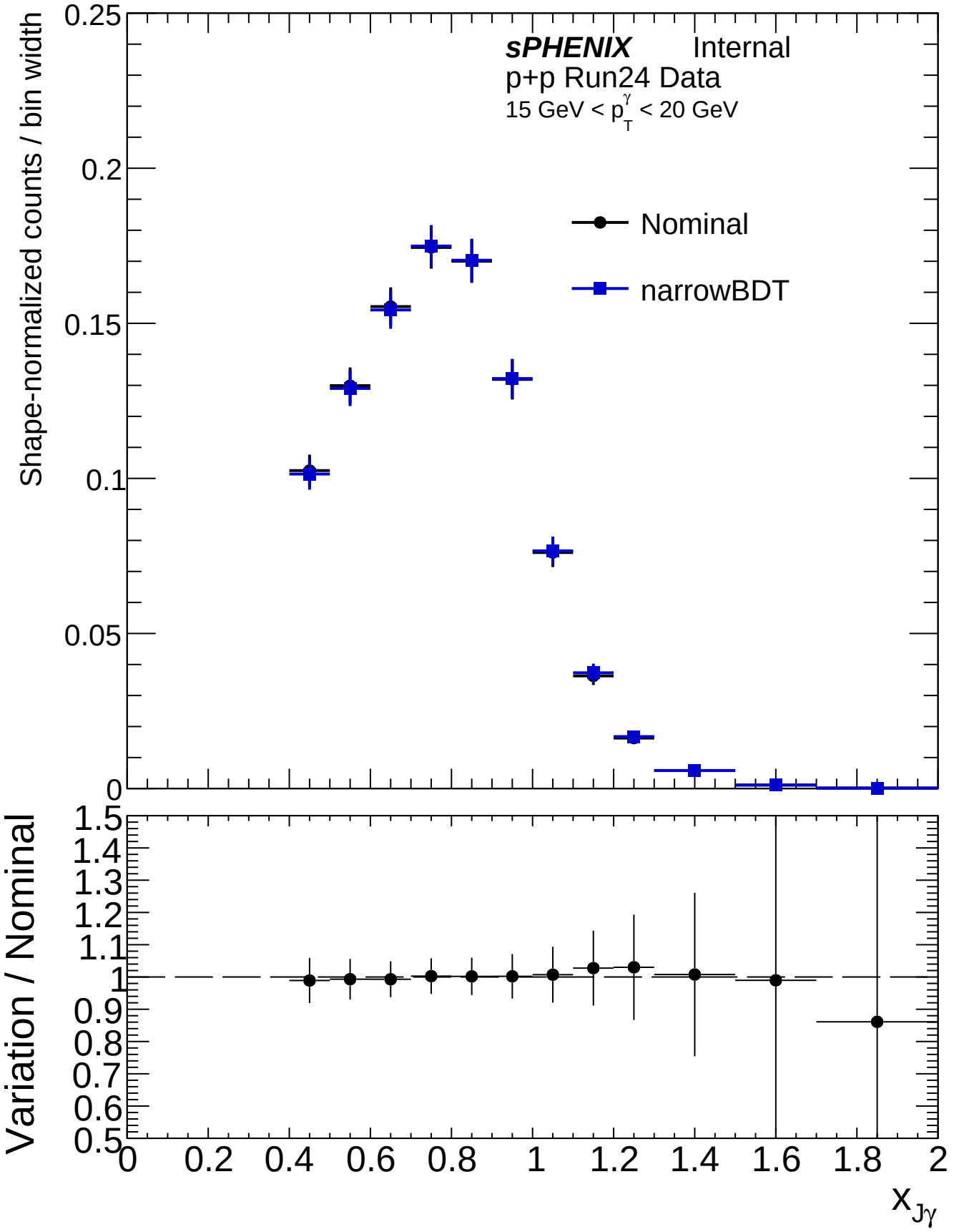
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
narrowBDT
Shape-normalized counts / bin width
0.25
0.2
0.15
0.1
0.05
0
Variation / Nominal
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
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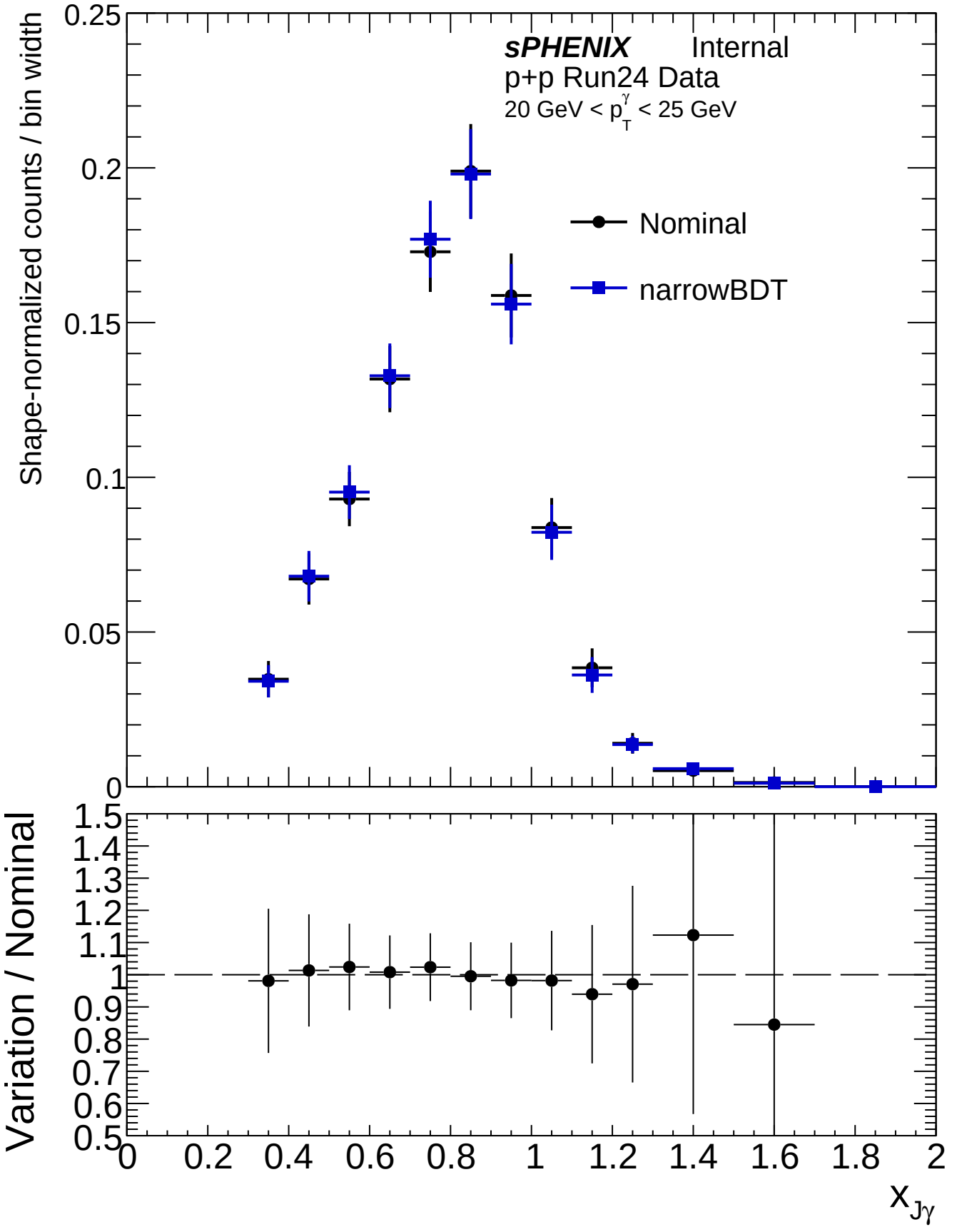
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
x_Jγ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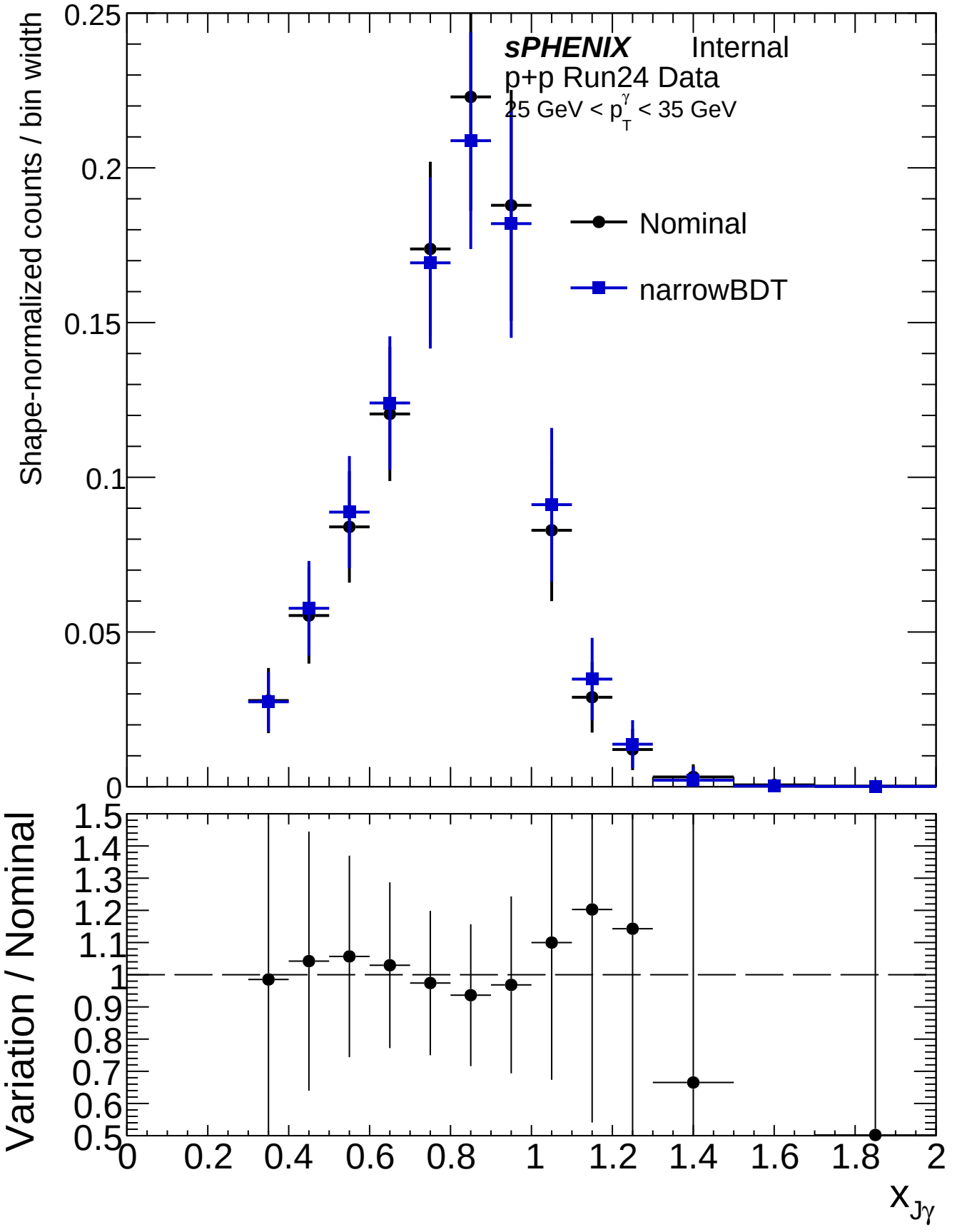
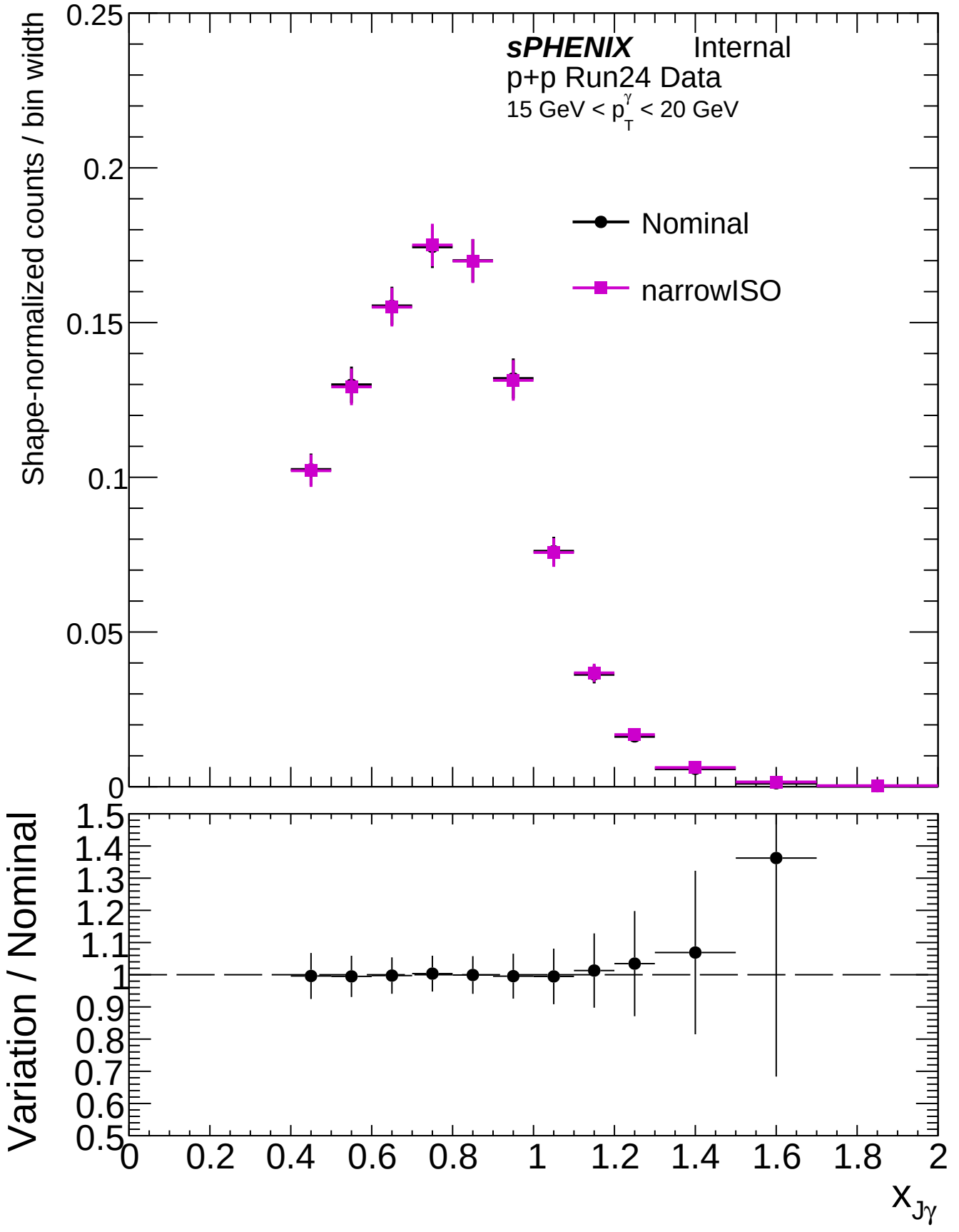
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
narrowBDT
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ


sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
narrowBDT
Shape-normalized counts / bin width
Variation / Nominal
$x_{J\gamma}$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p+p Run24 Data
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Nominal
narrowISO
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
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0.8
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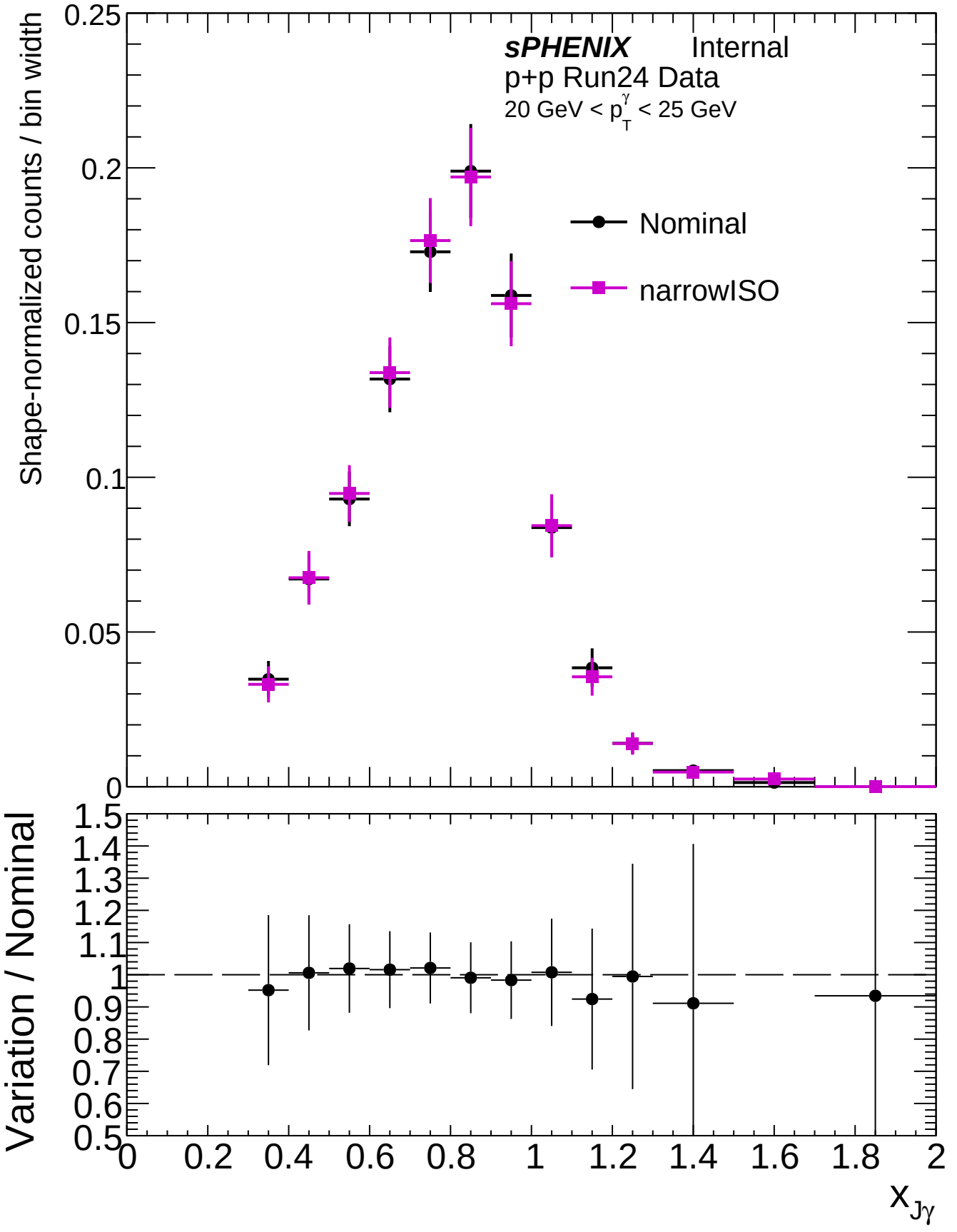
sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
narrowISO
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
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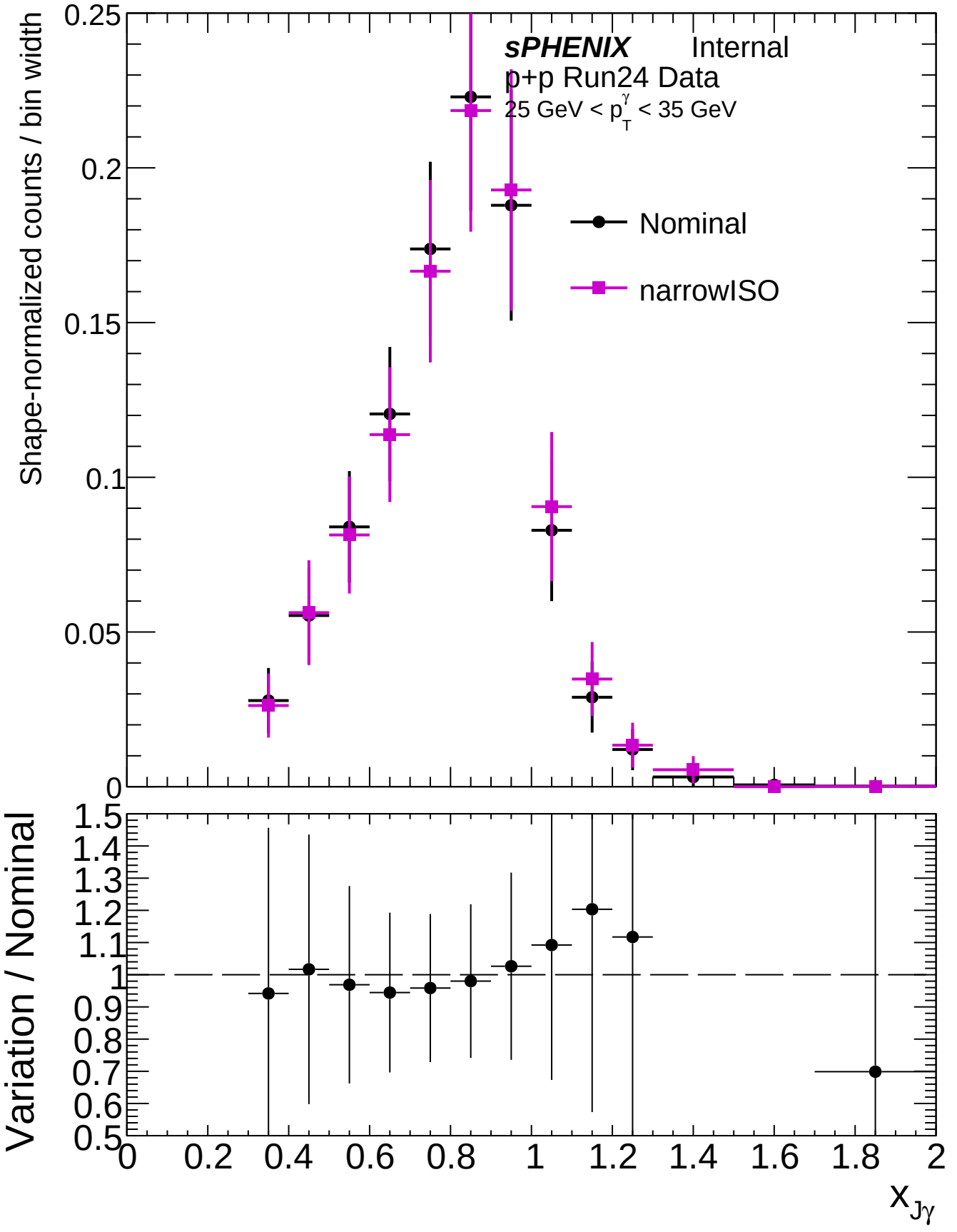
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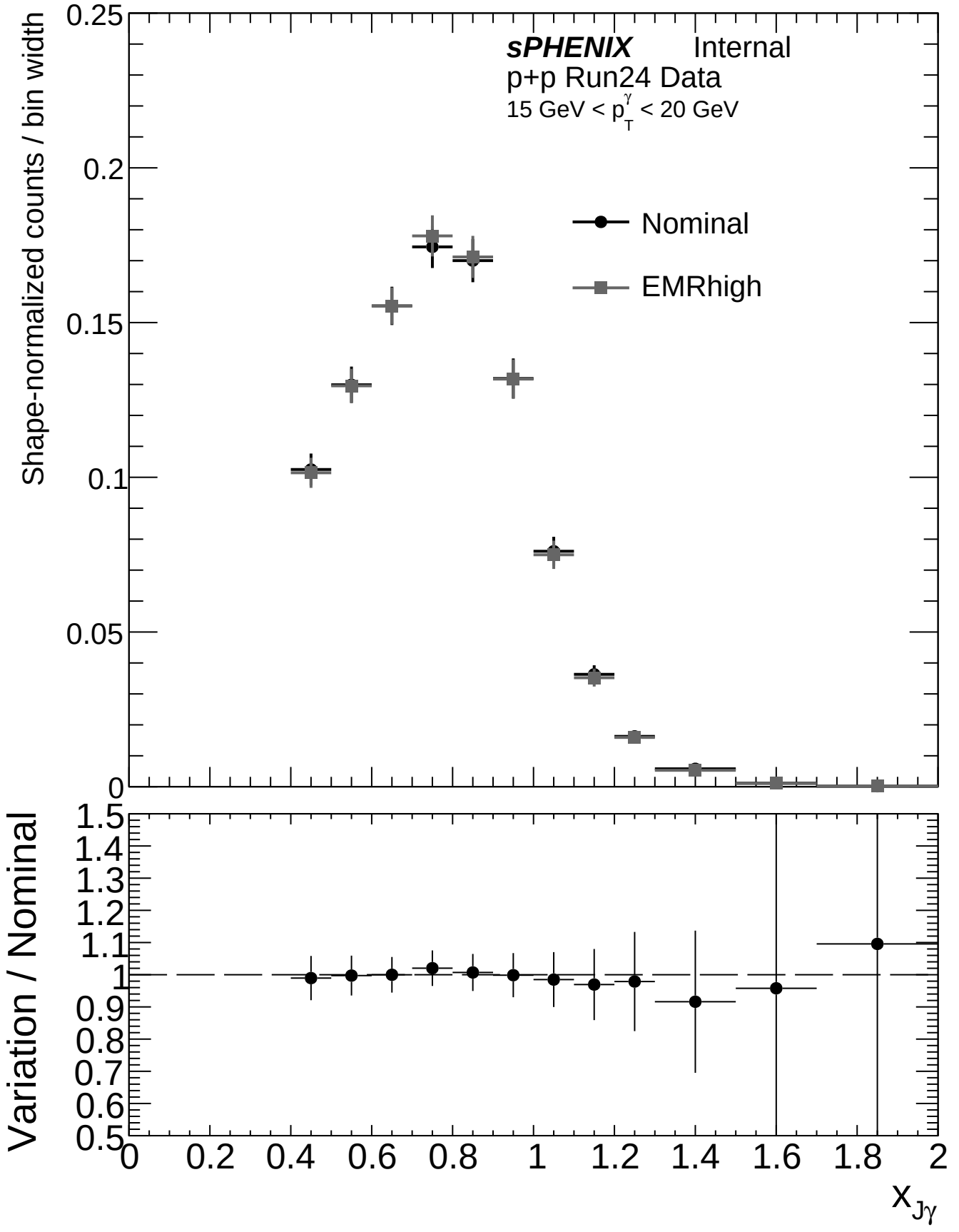
25 GeV < p_T^γ < 35 GeV
Nominal
narrowISO
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ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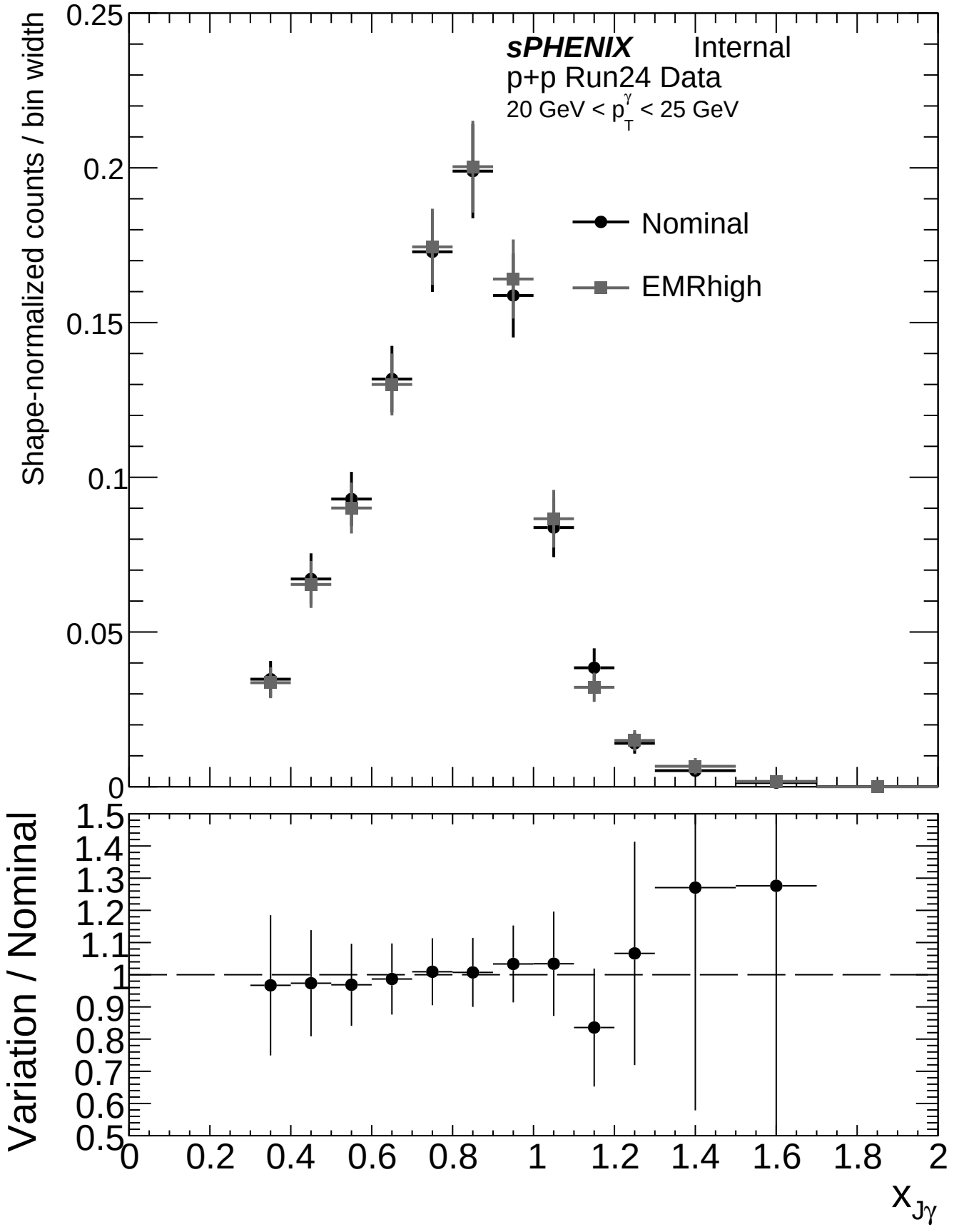
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
EMRhigh
Shape-normalized counts / bin width
Variation / Nominal
$x_{J\gamma}$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p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
EMRhigh
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
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1.5
1.4
1.3
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0.8
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1.6
1.8
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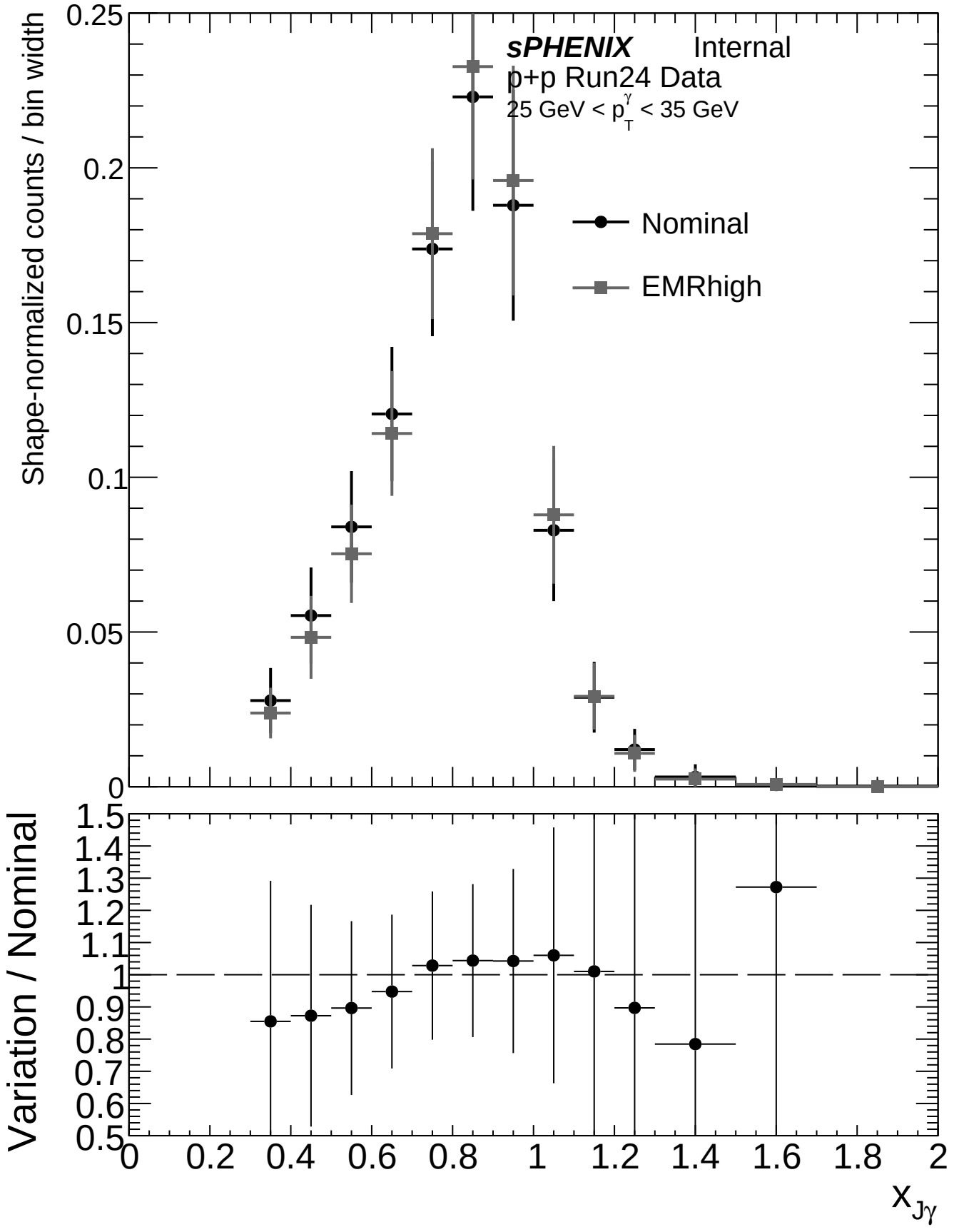
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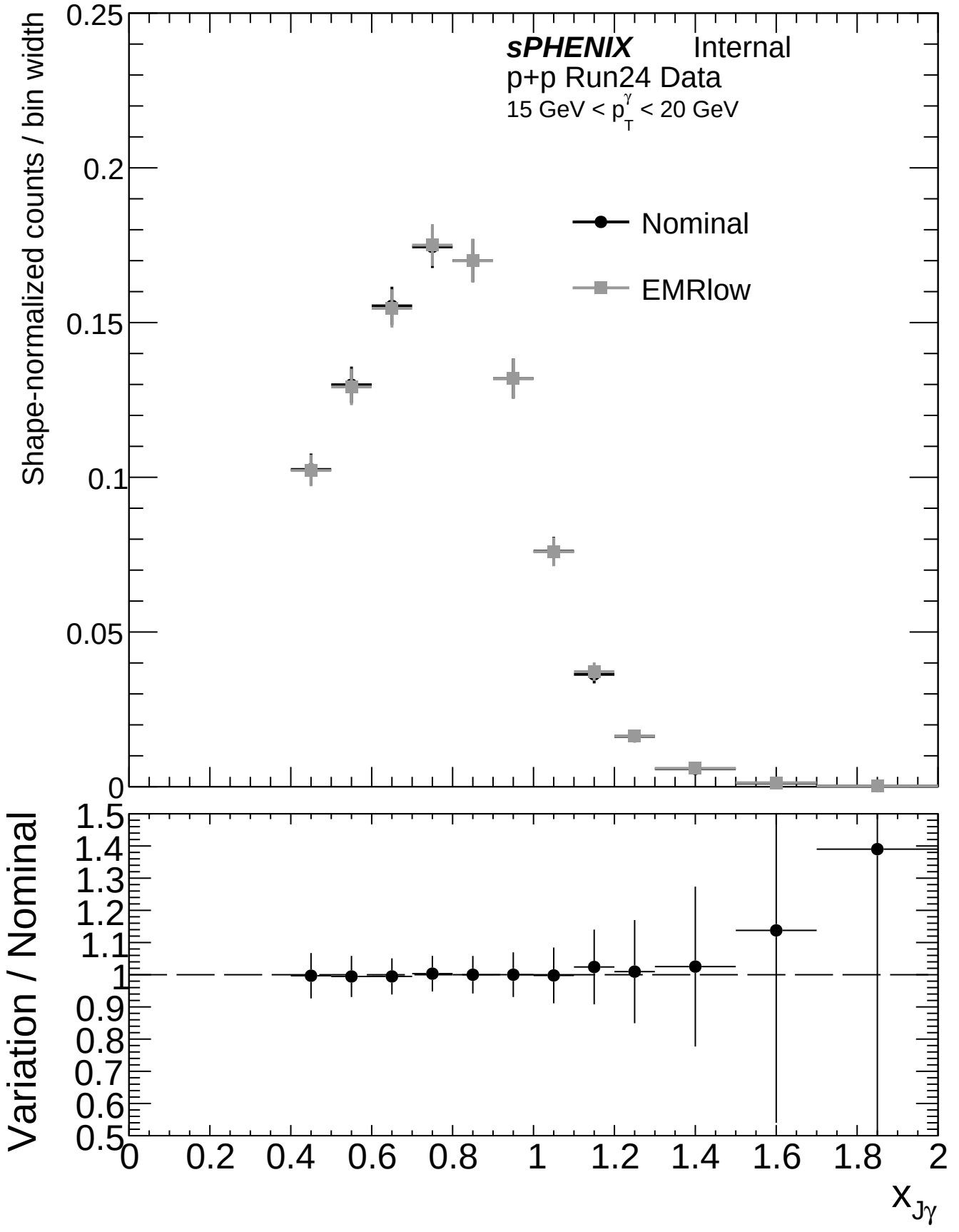
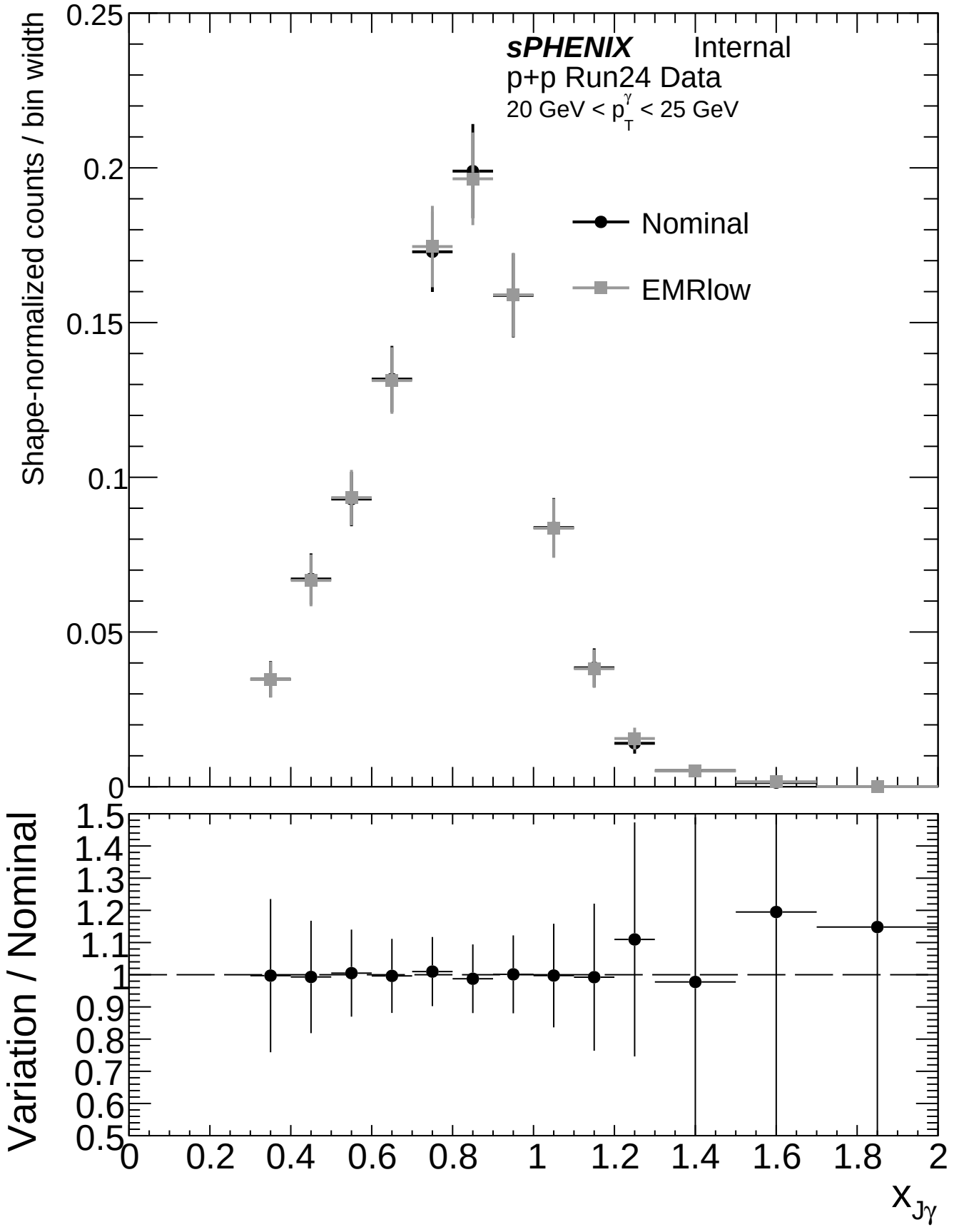
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
EMRhigh
Shape-normalized counts / bin width
Variation / Nominal
$x_{J\gamma}$


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
EMRlow
Shape-normalized counts / bin width
Variation / Nominal
$x_{J\gamma}$


sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
EMRlow
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
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1.5
1.4
1.3
1.2
1.1
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0.9
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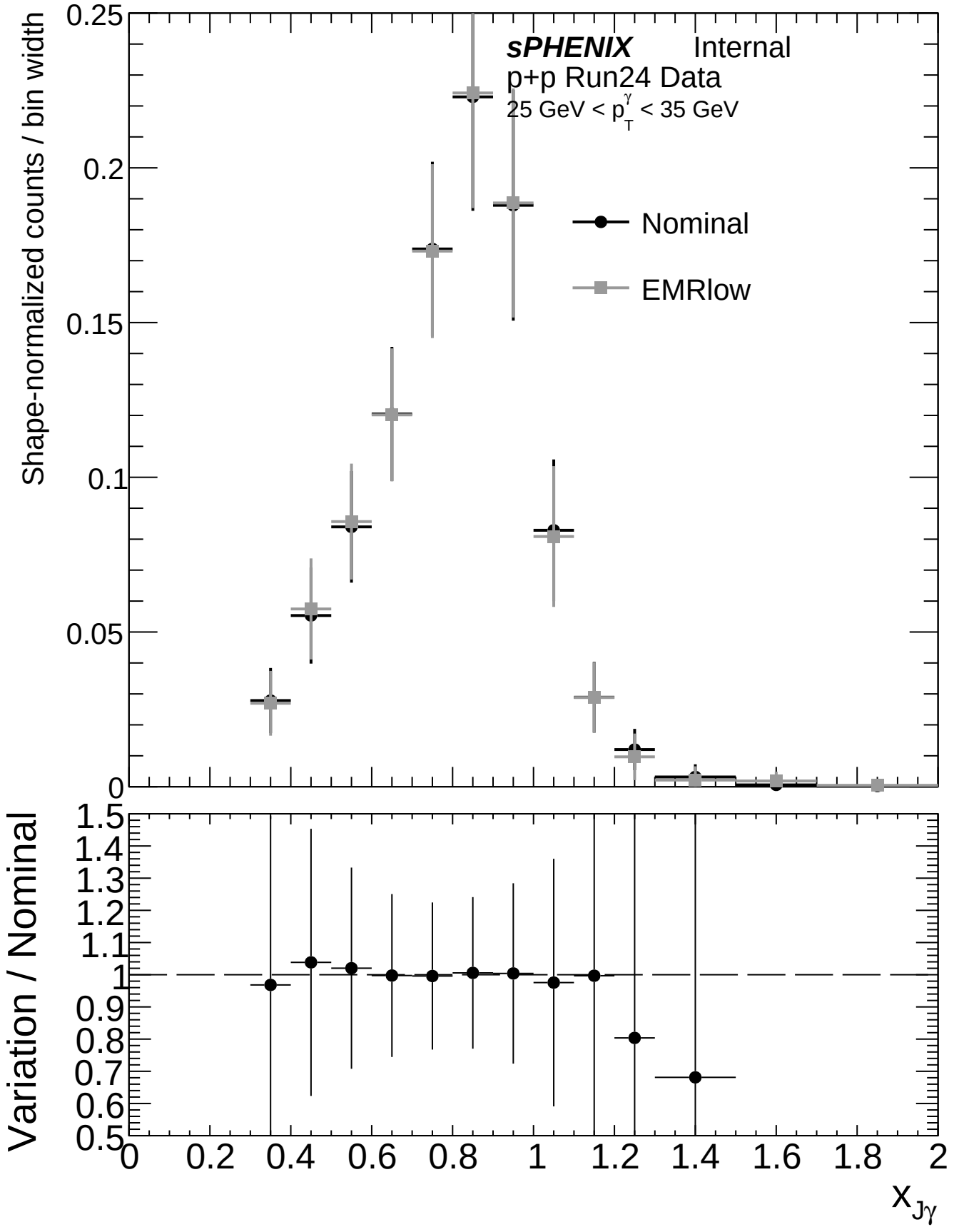
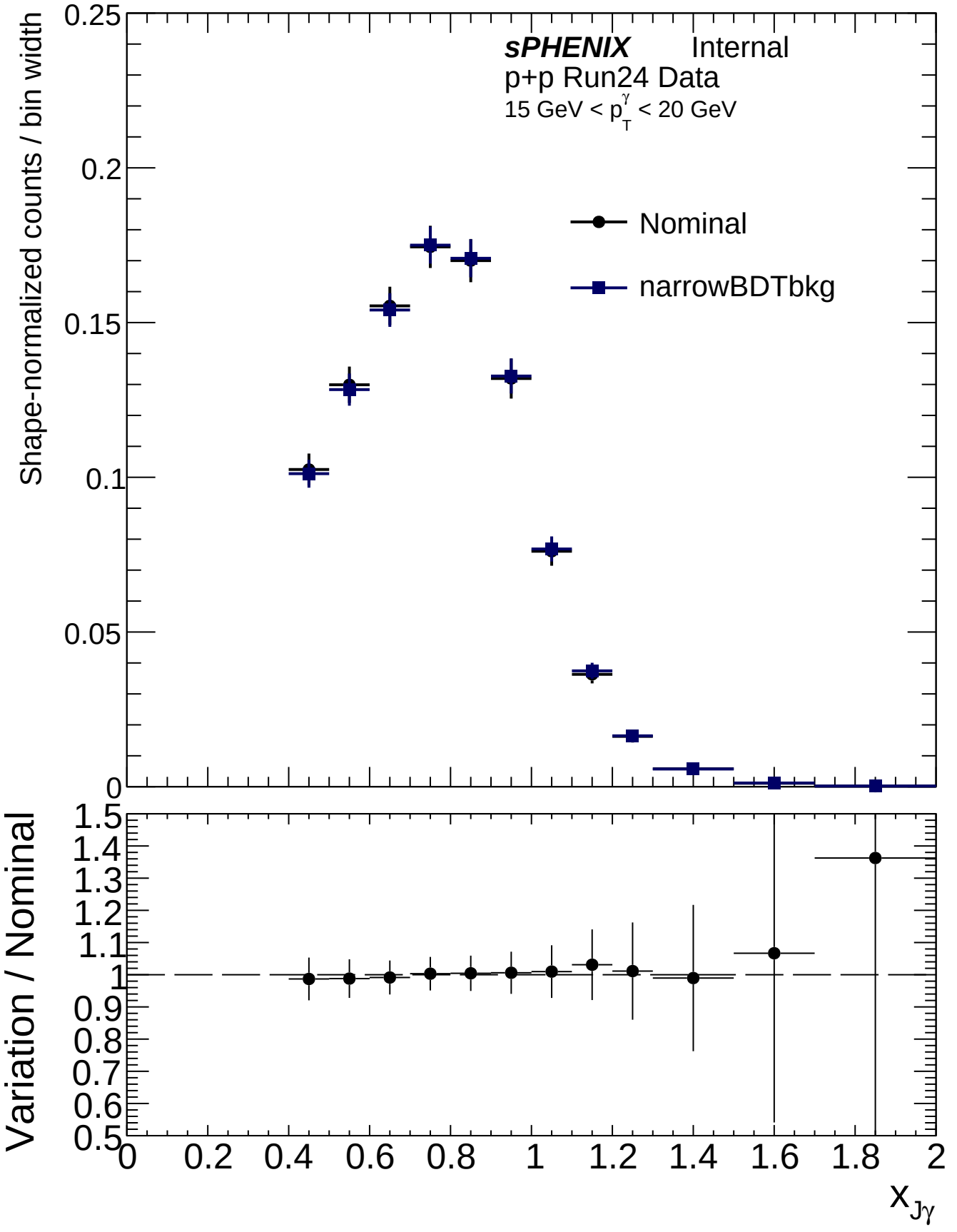
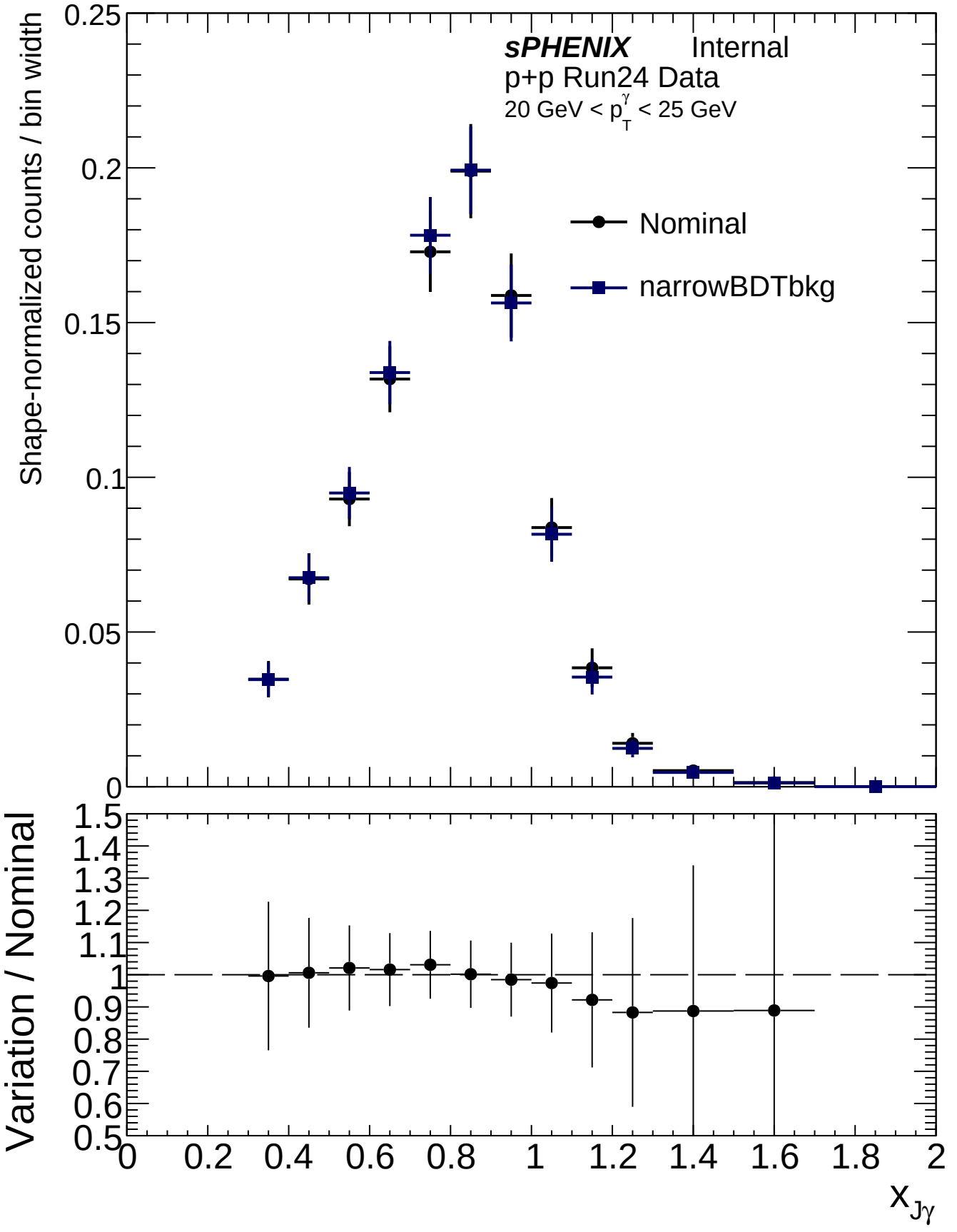
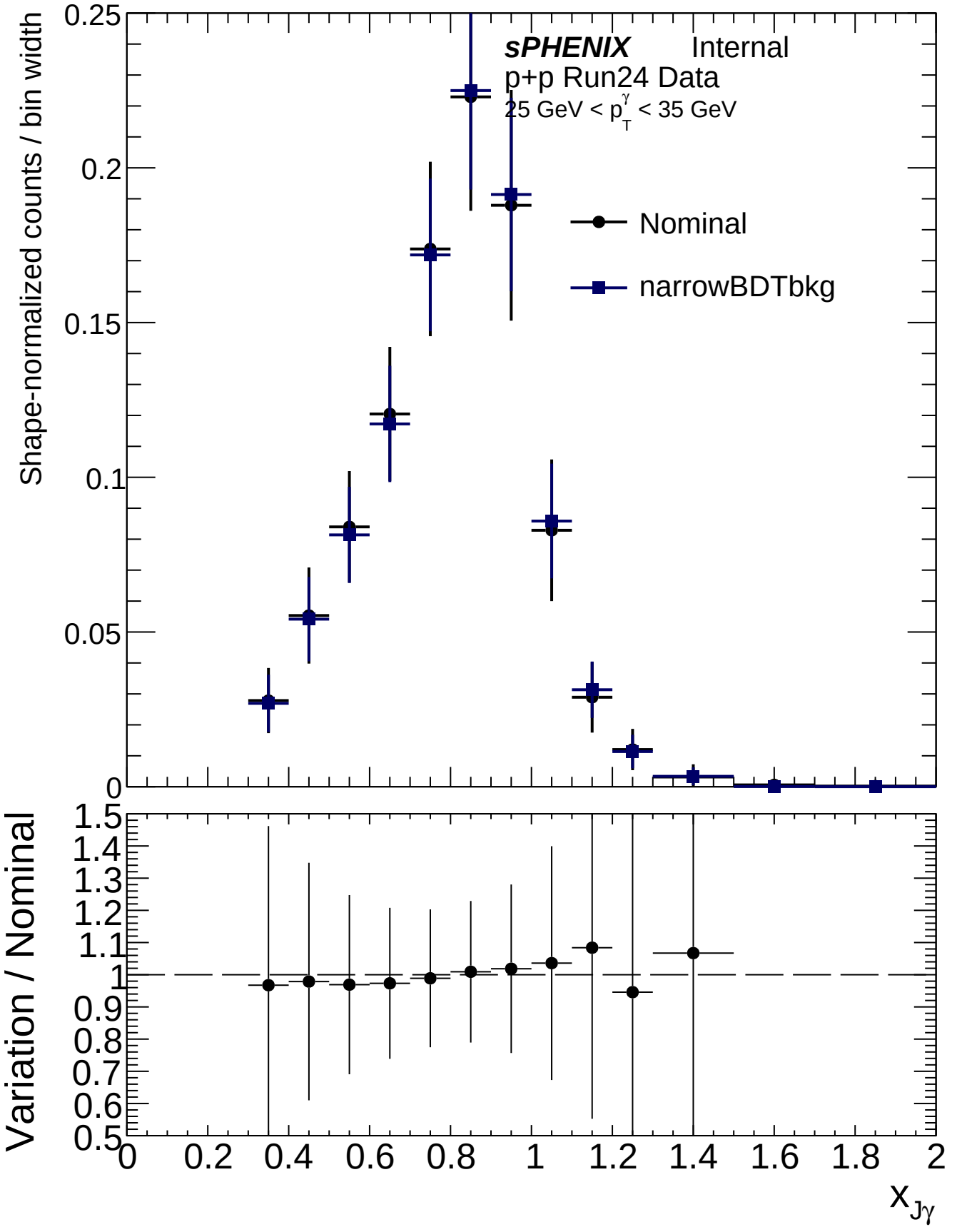
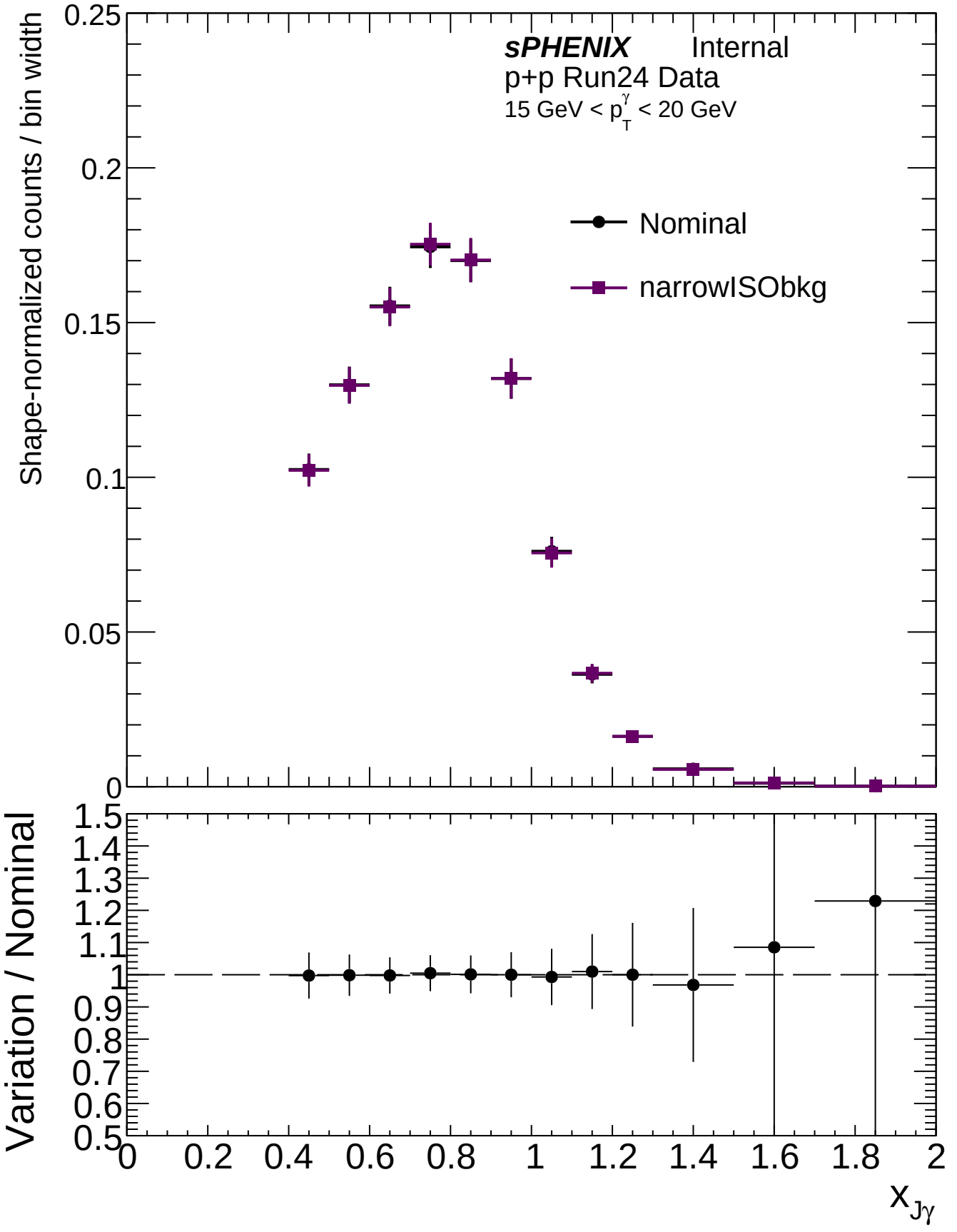
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

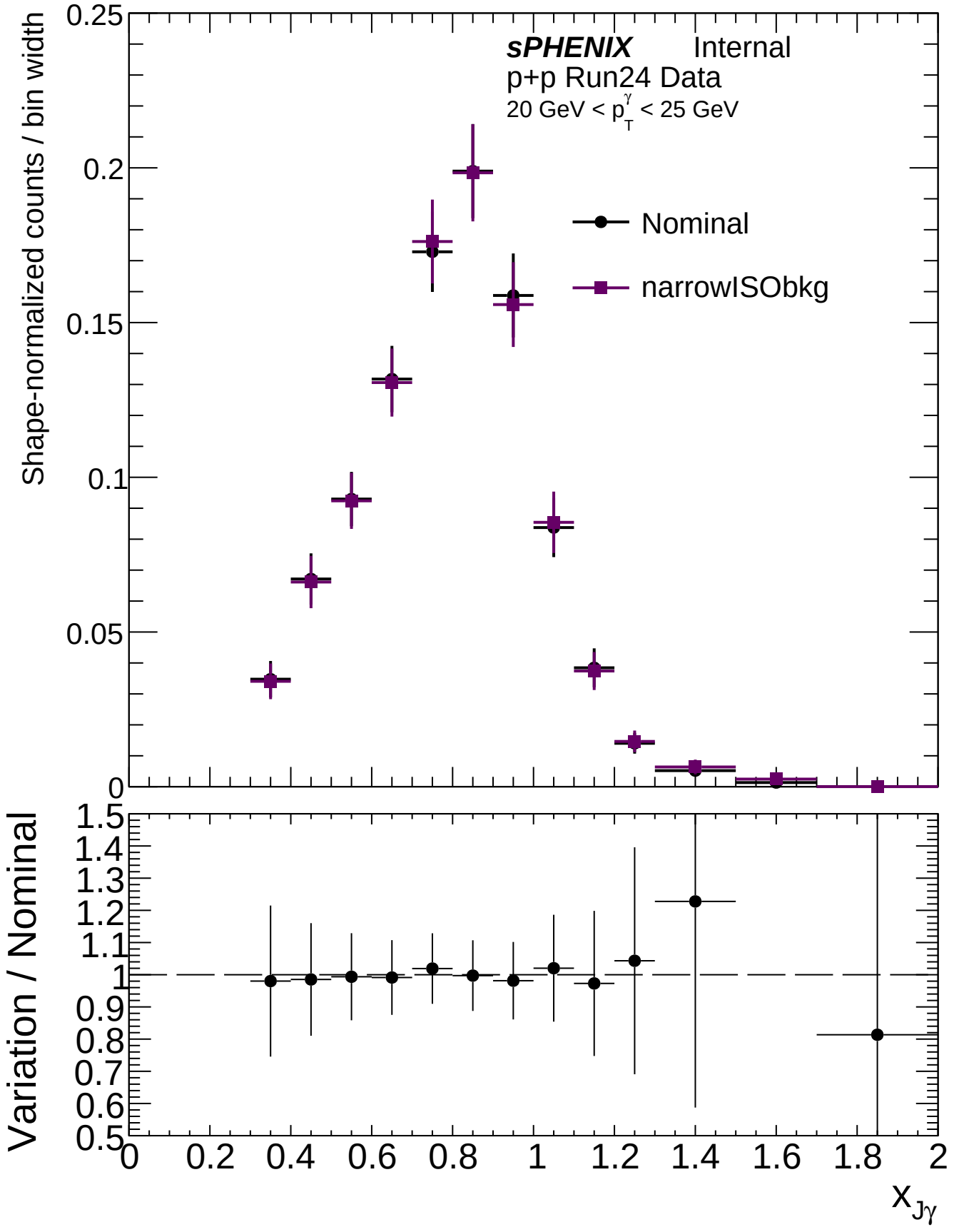

sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
EMRlow
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

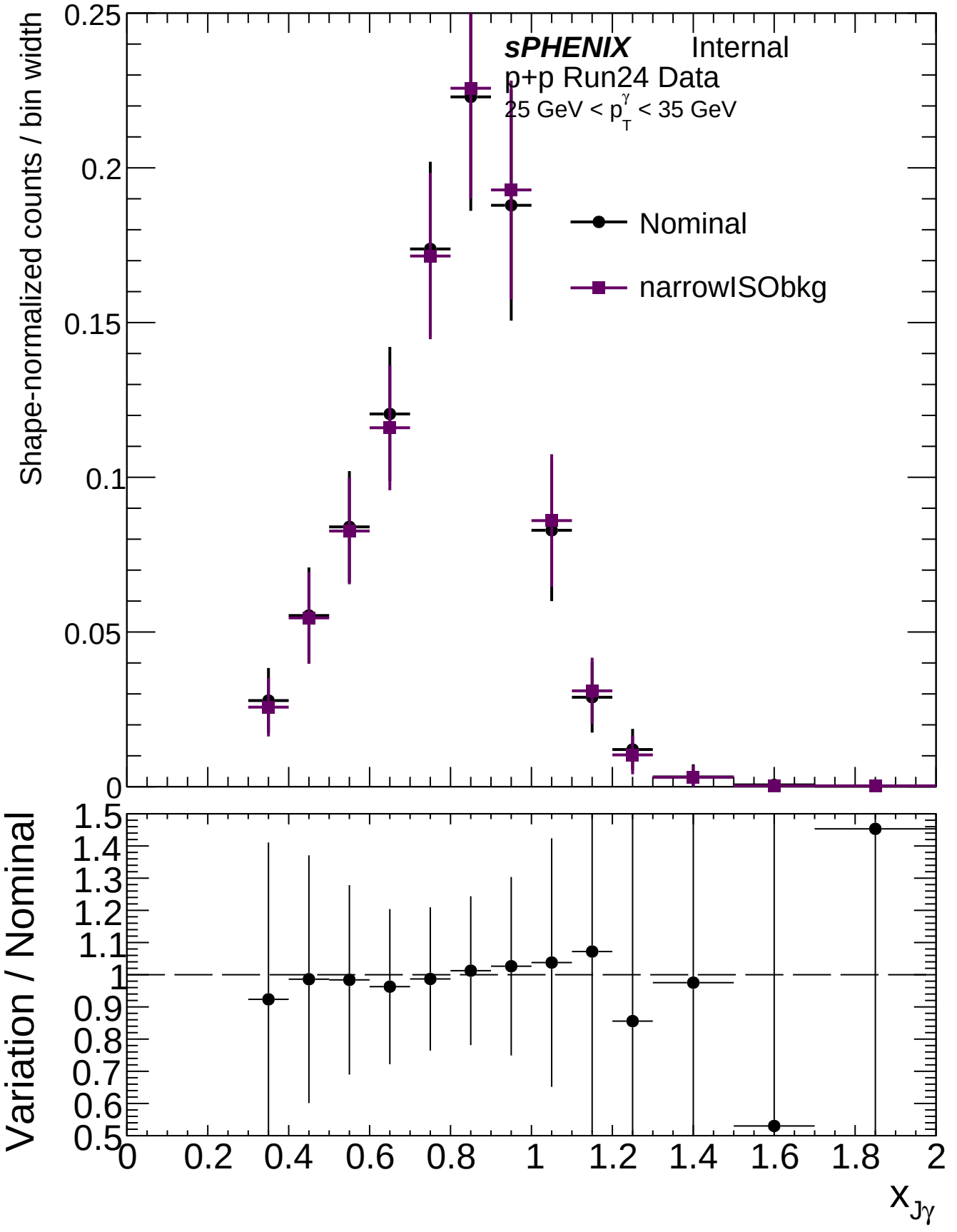

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
narrowBDTbkg
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

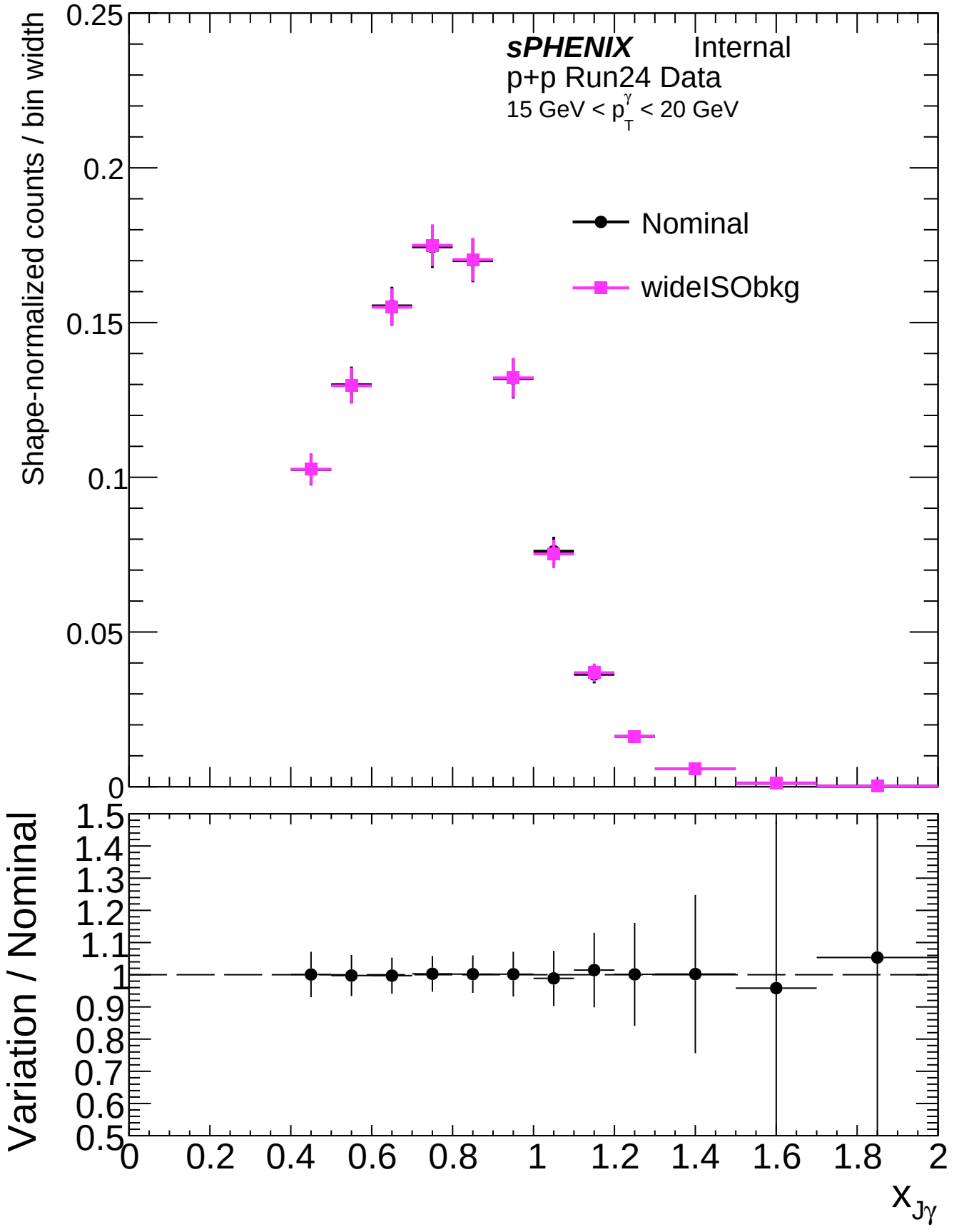

sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
narrowBDTbkg
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

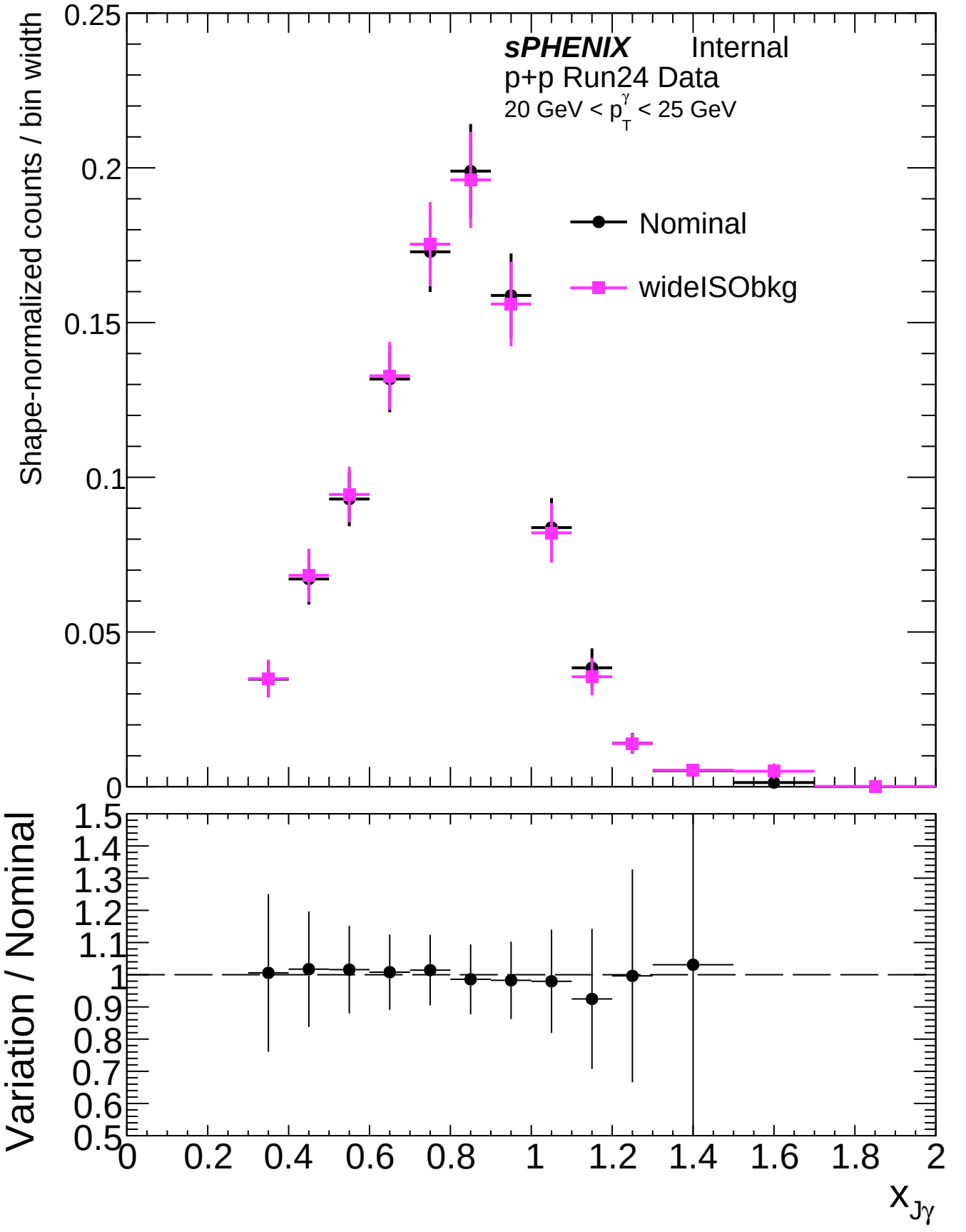

sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
narrowBDTbkg
Shape-normalized counts / bin width
Variation / Nominal
$x_{J\gamma}$
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

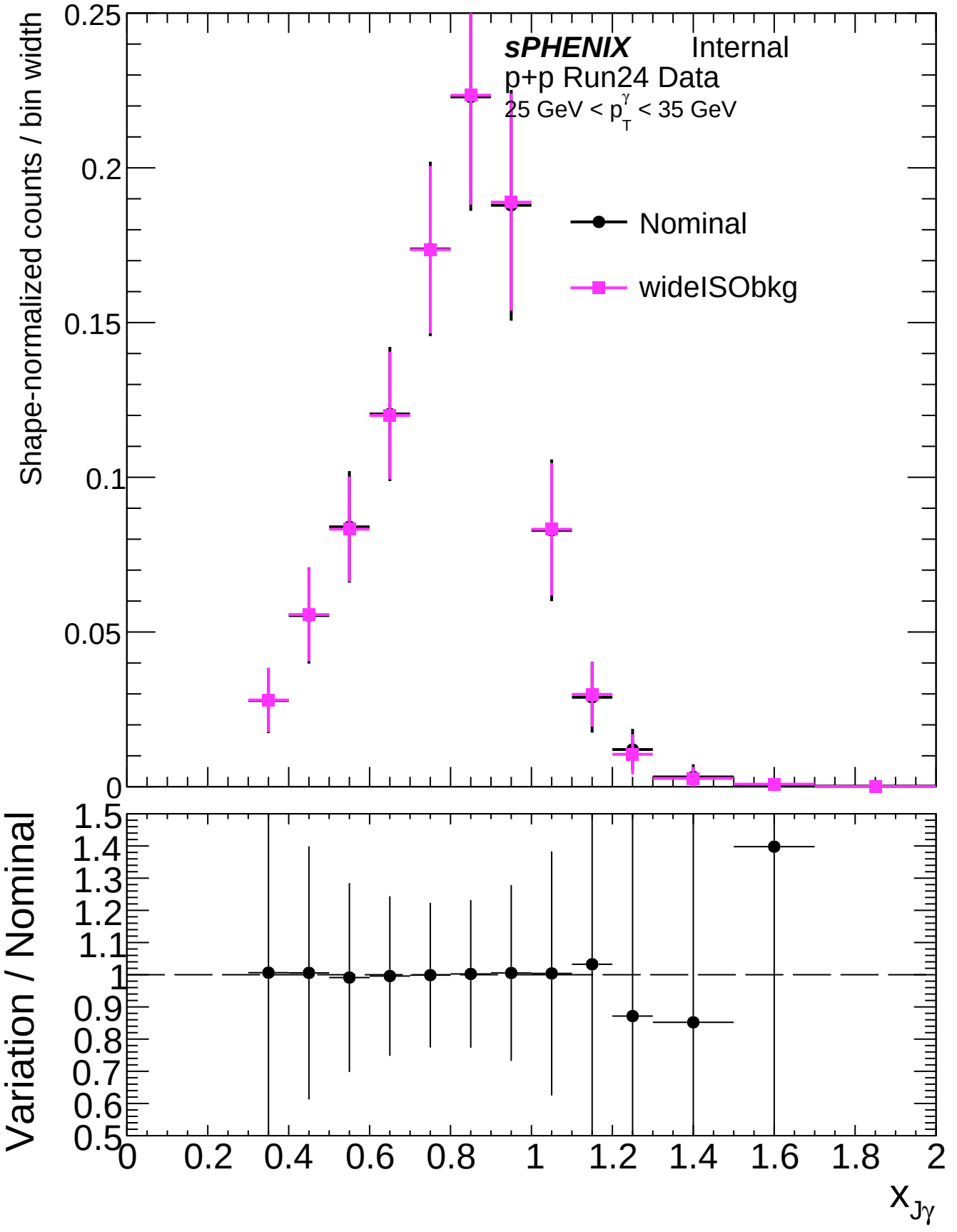

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
narrowISObkg
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ

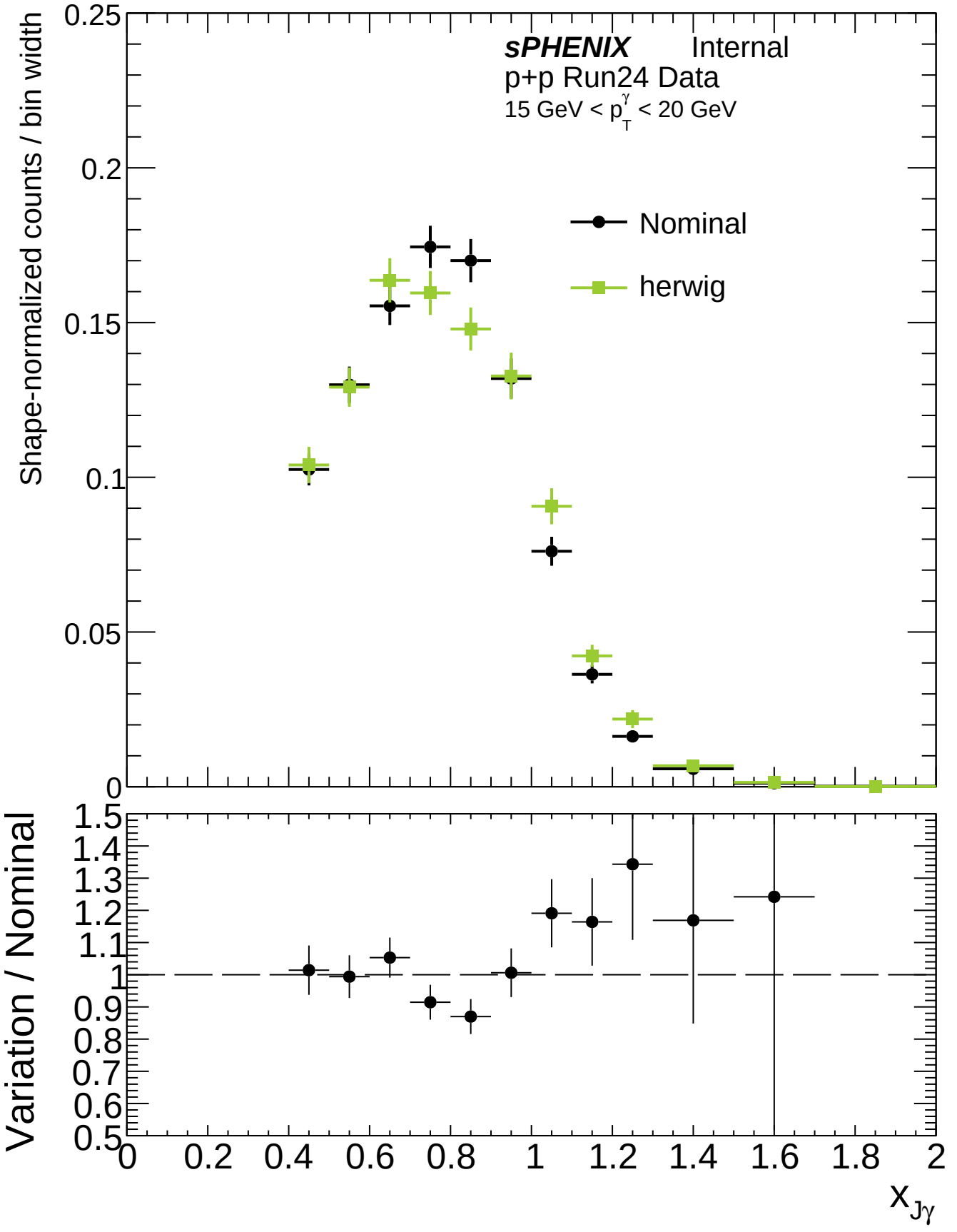

sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
narrowISObkg
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

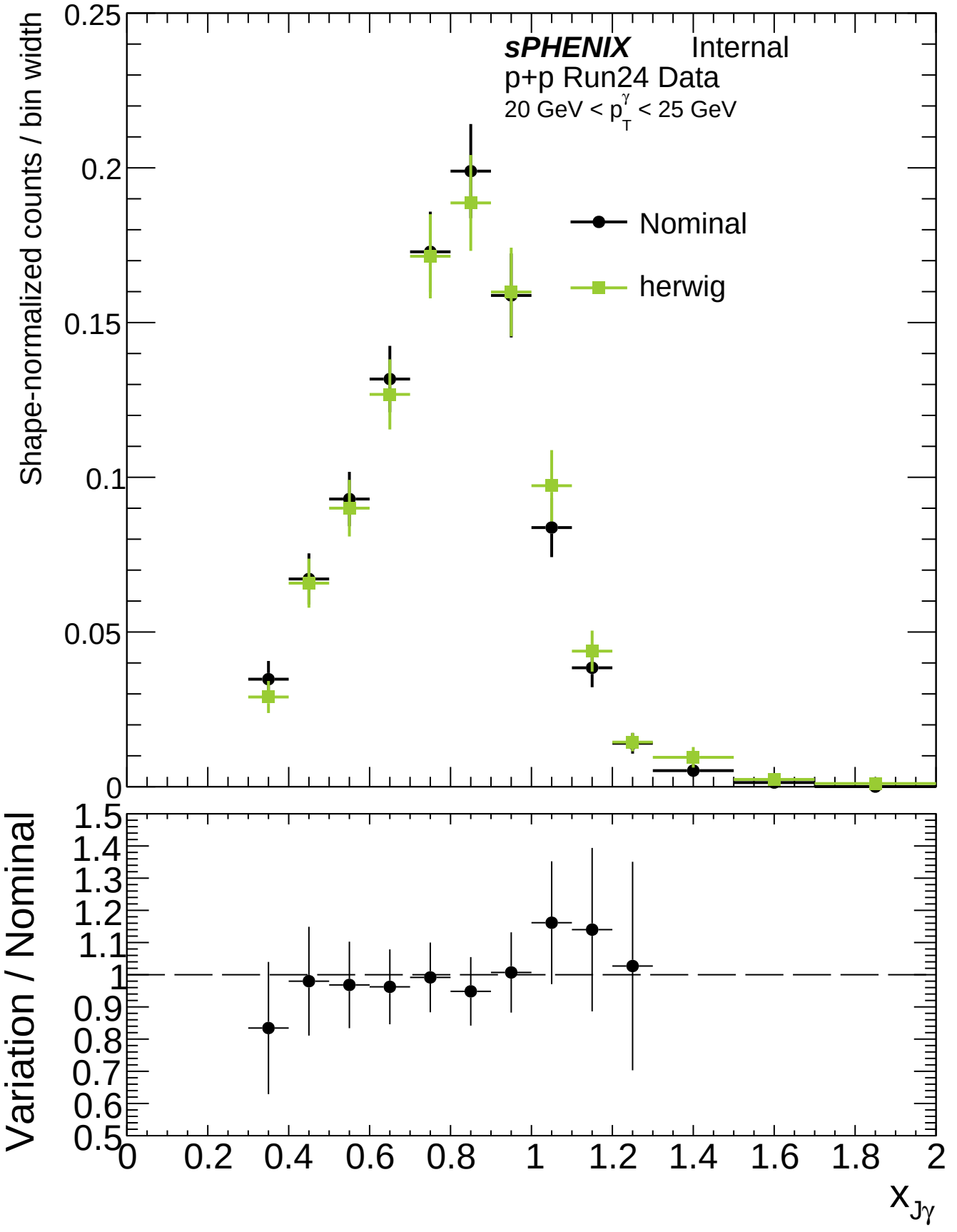

sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
narrowISObkg
Shape-normalized counts / bin width
Variation / Nominal
$x_{J\gamma}$

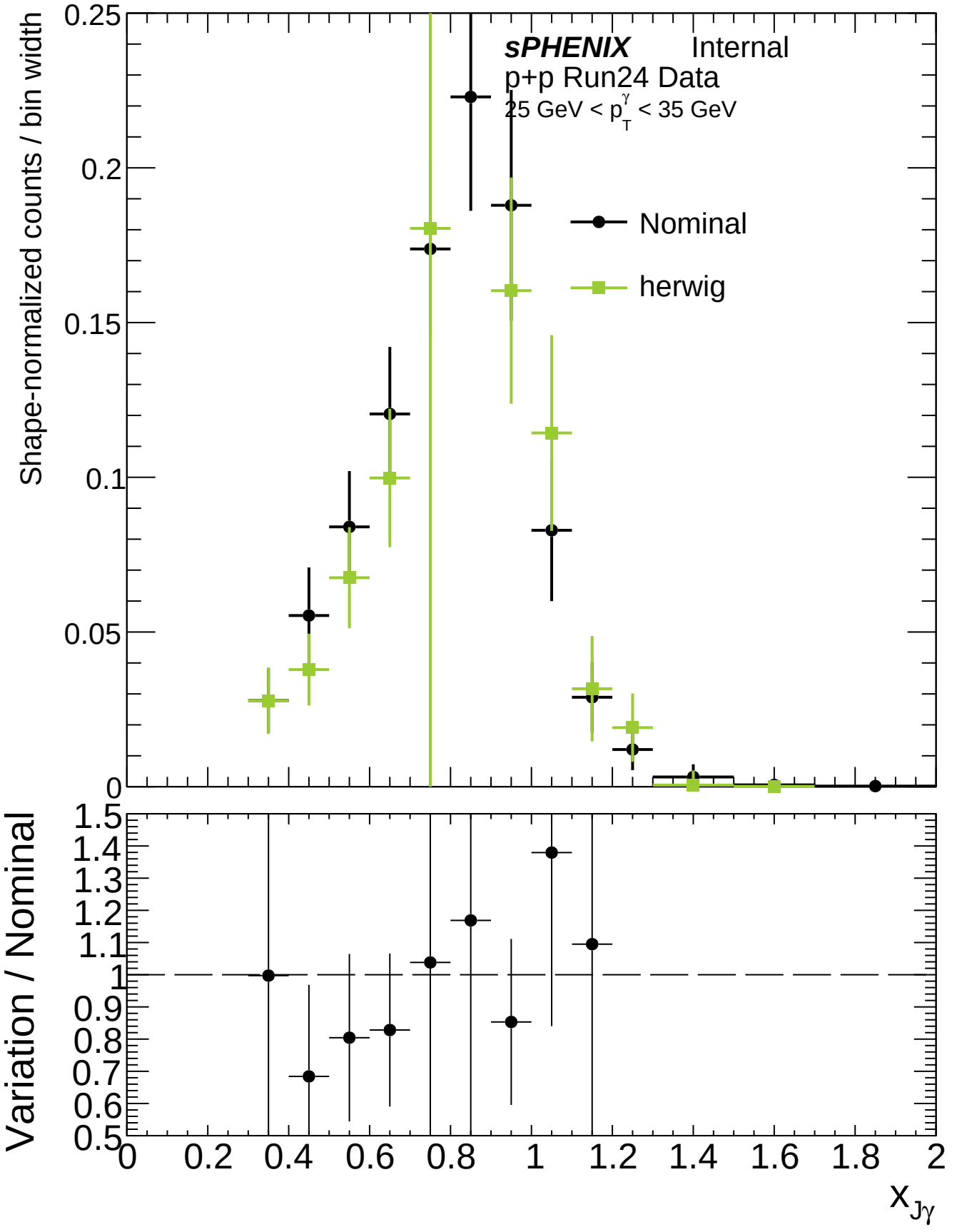

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
wideISObkg
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ

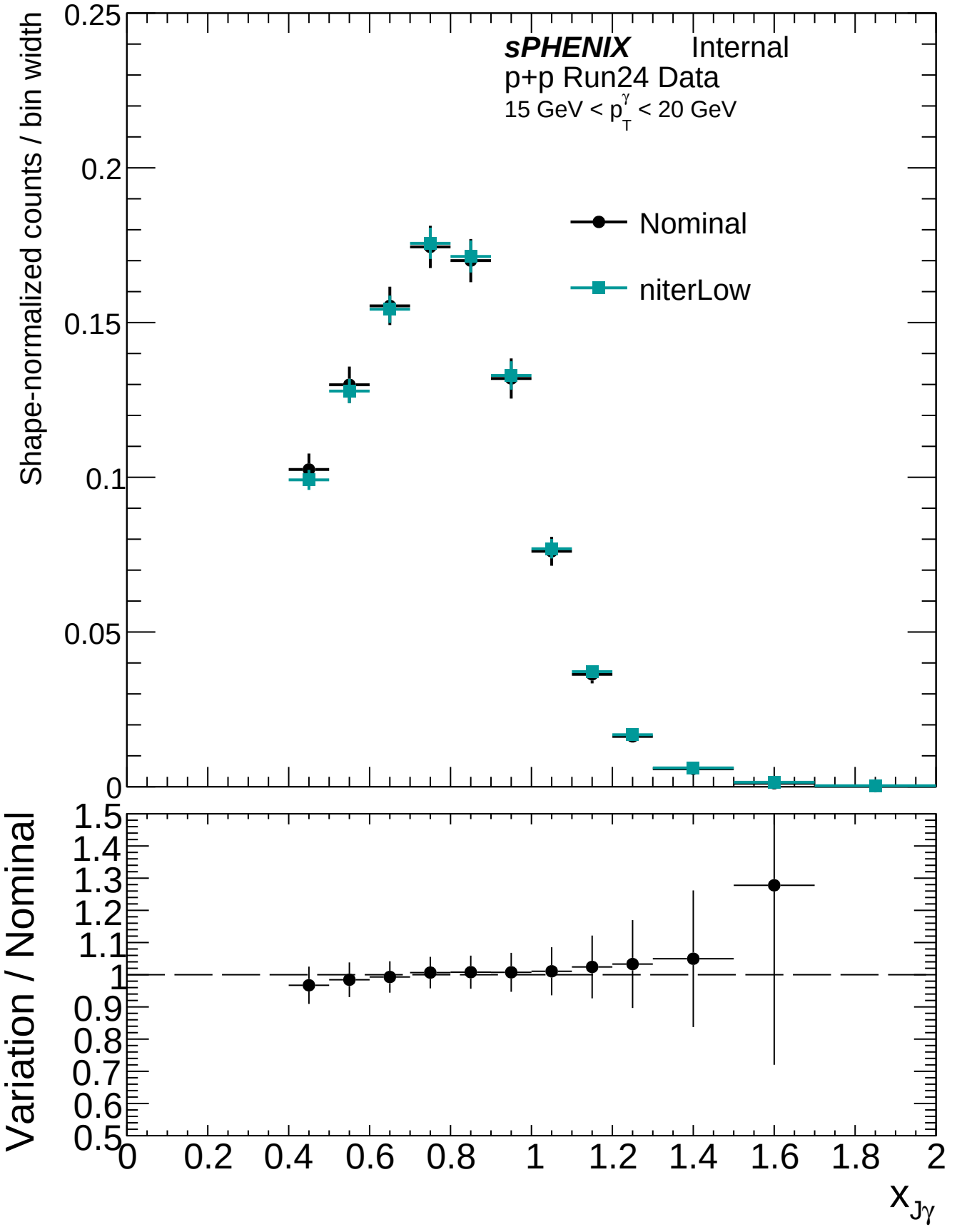

sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
wideISObkg
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ

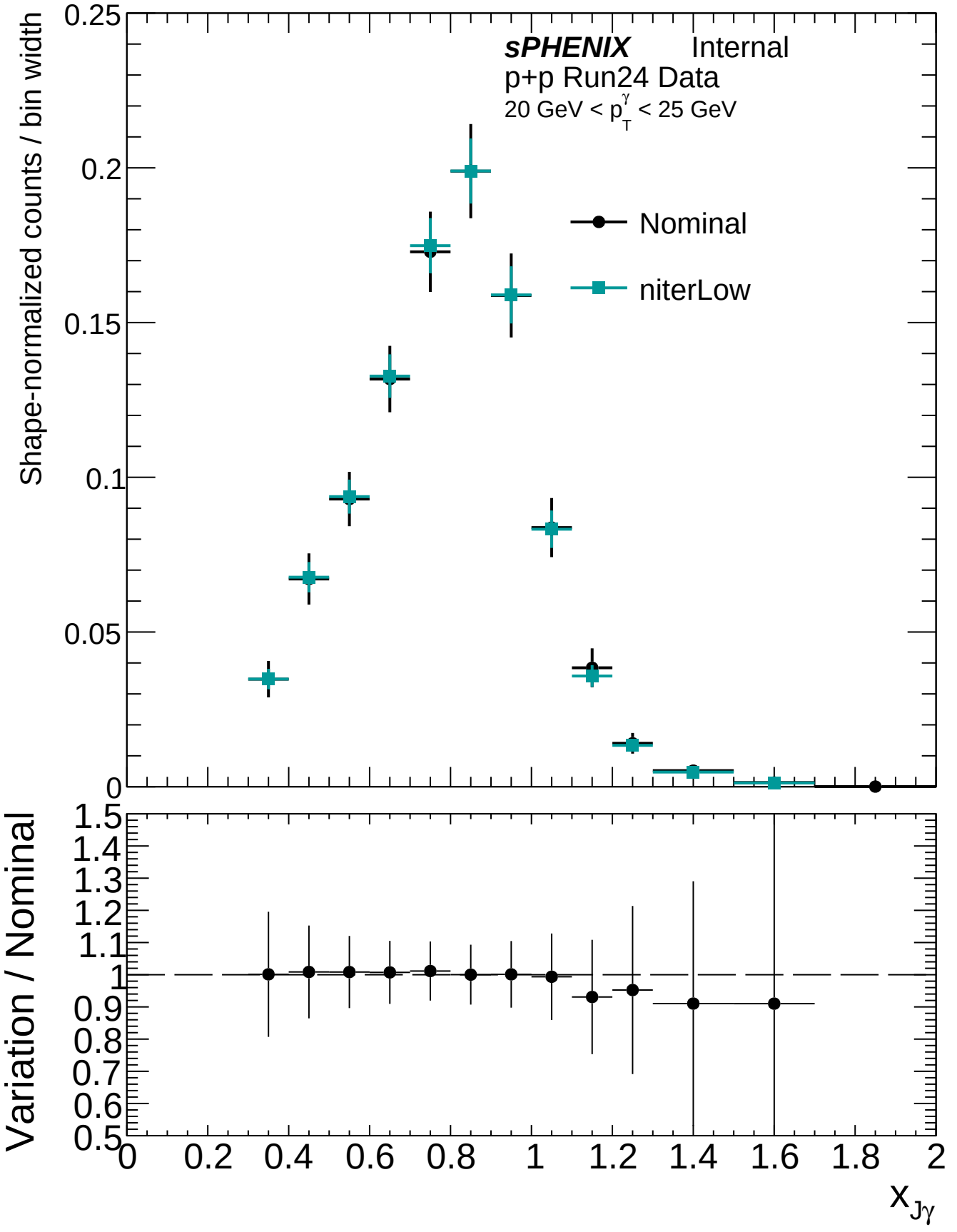

sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
wideISObkg
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

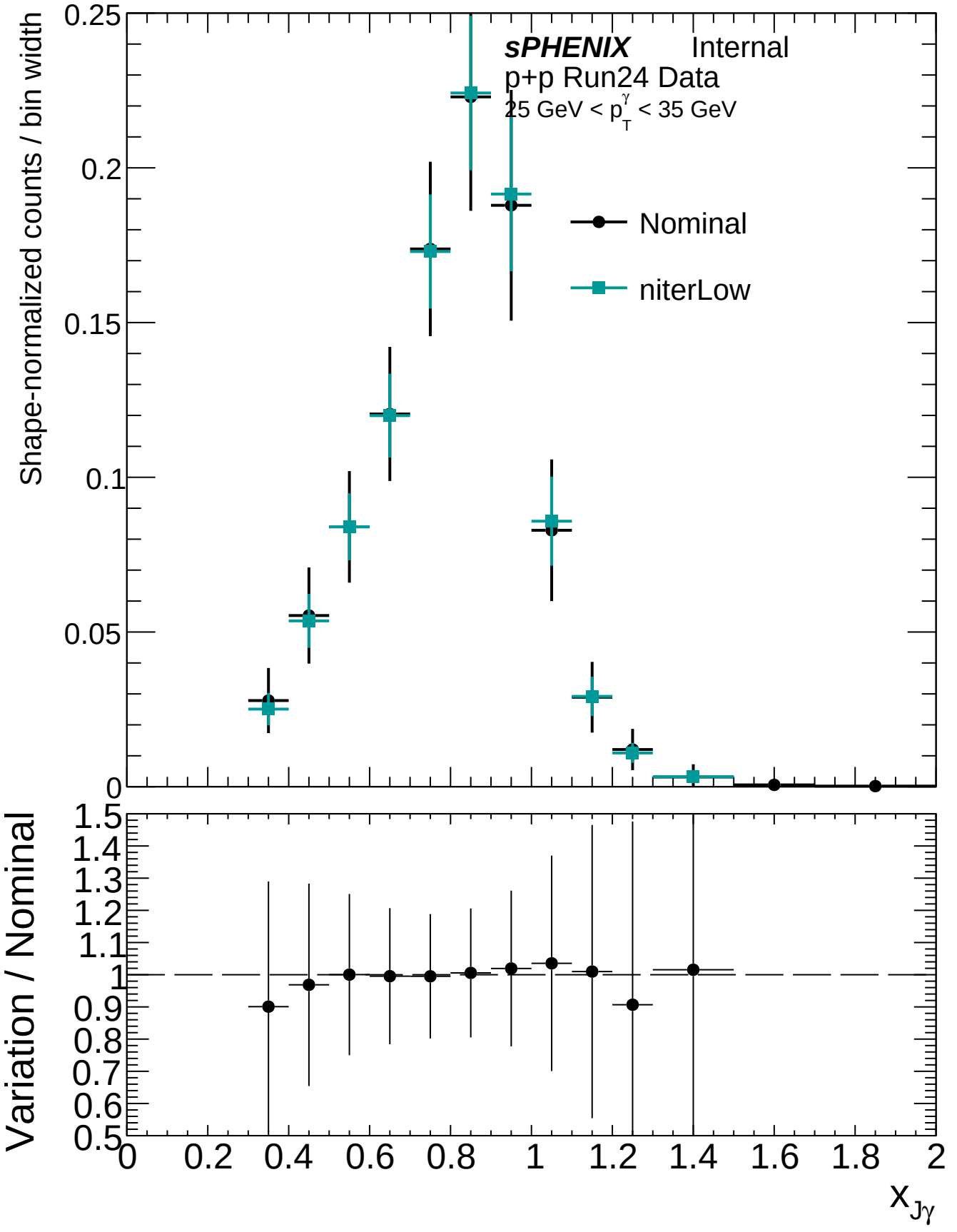

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
herwig
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

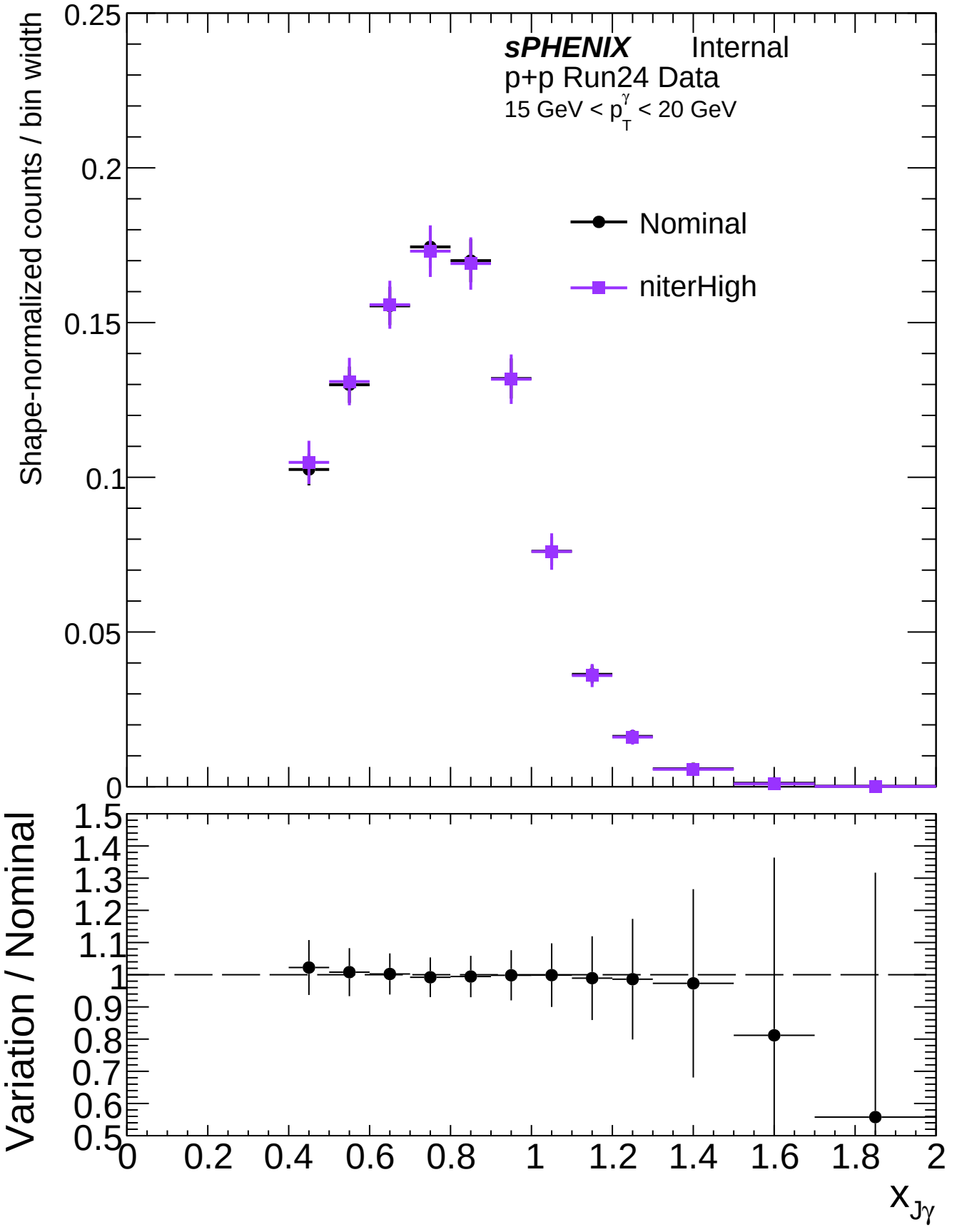

sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
herwig
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ

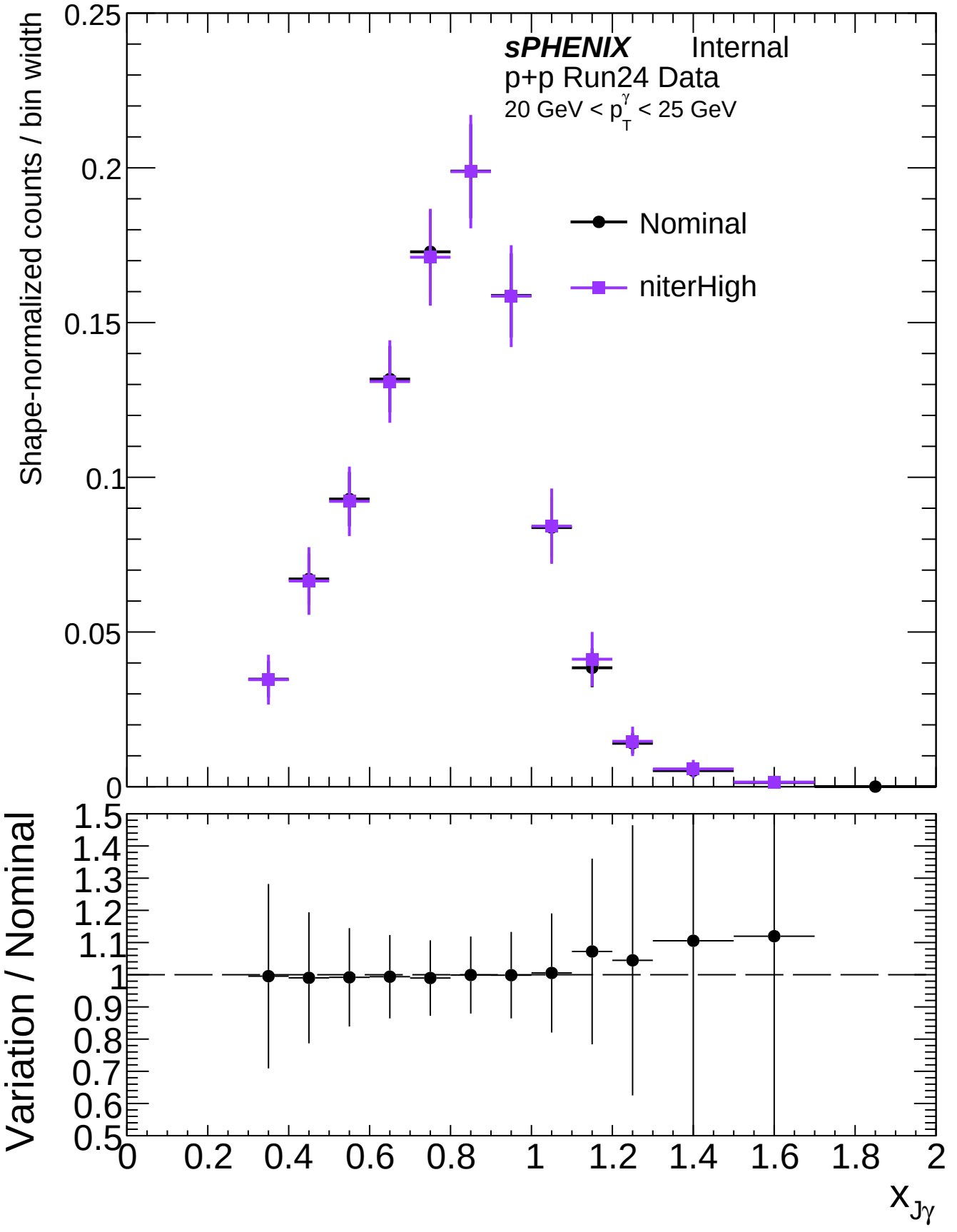

sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
herwig
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

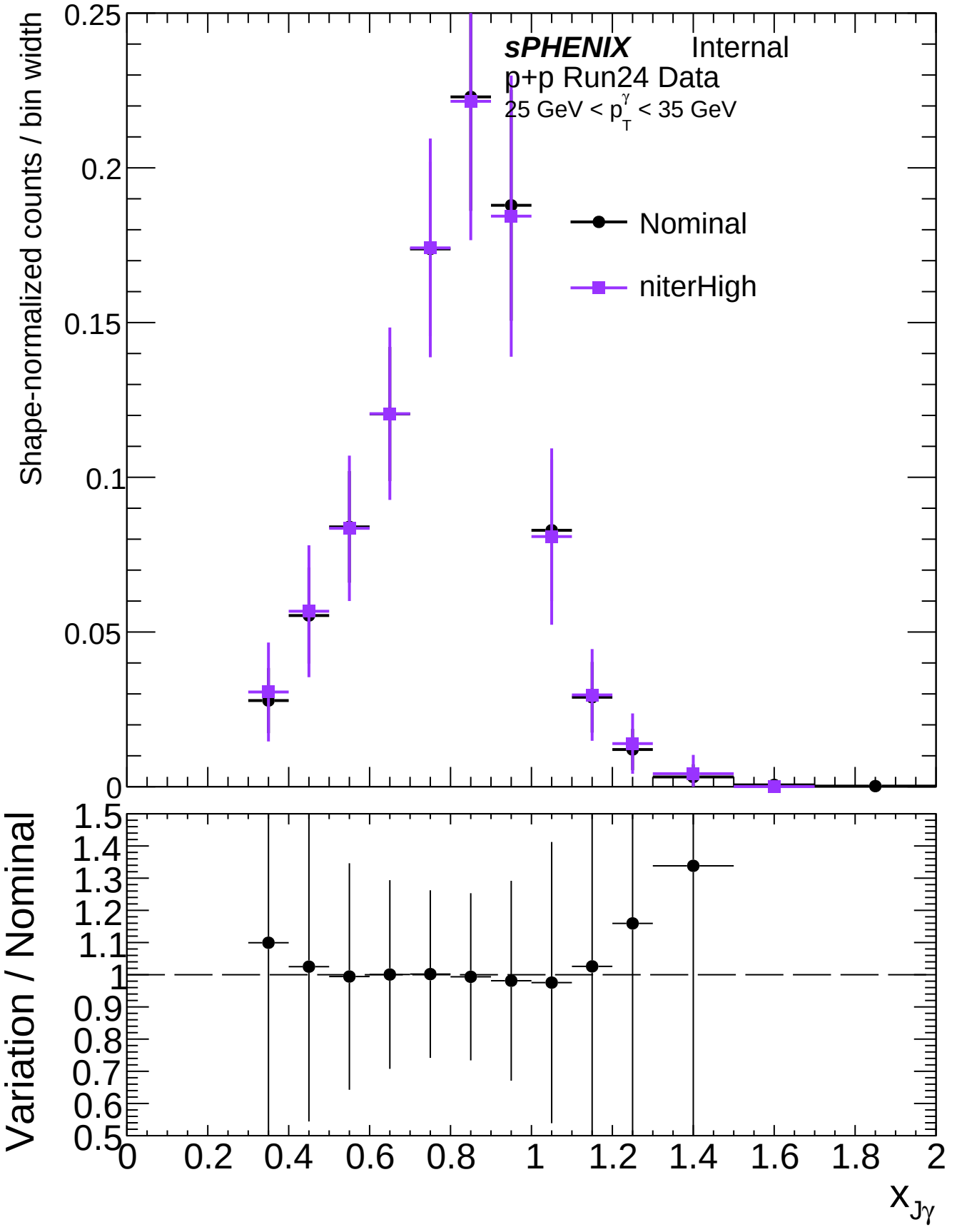

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
niterLow
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

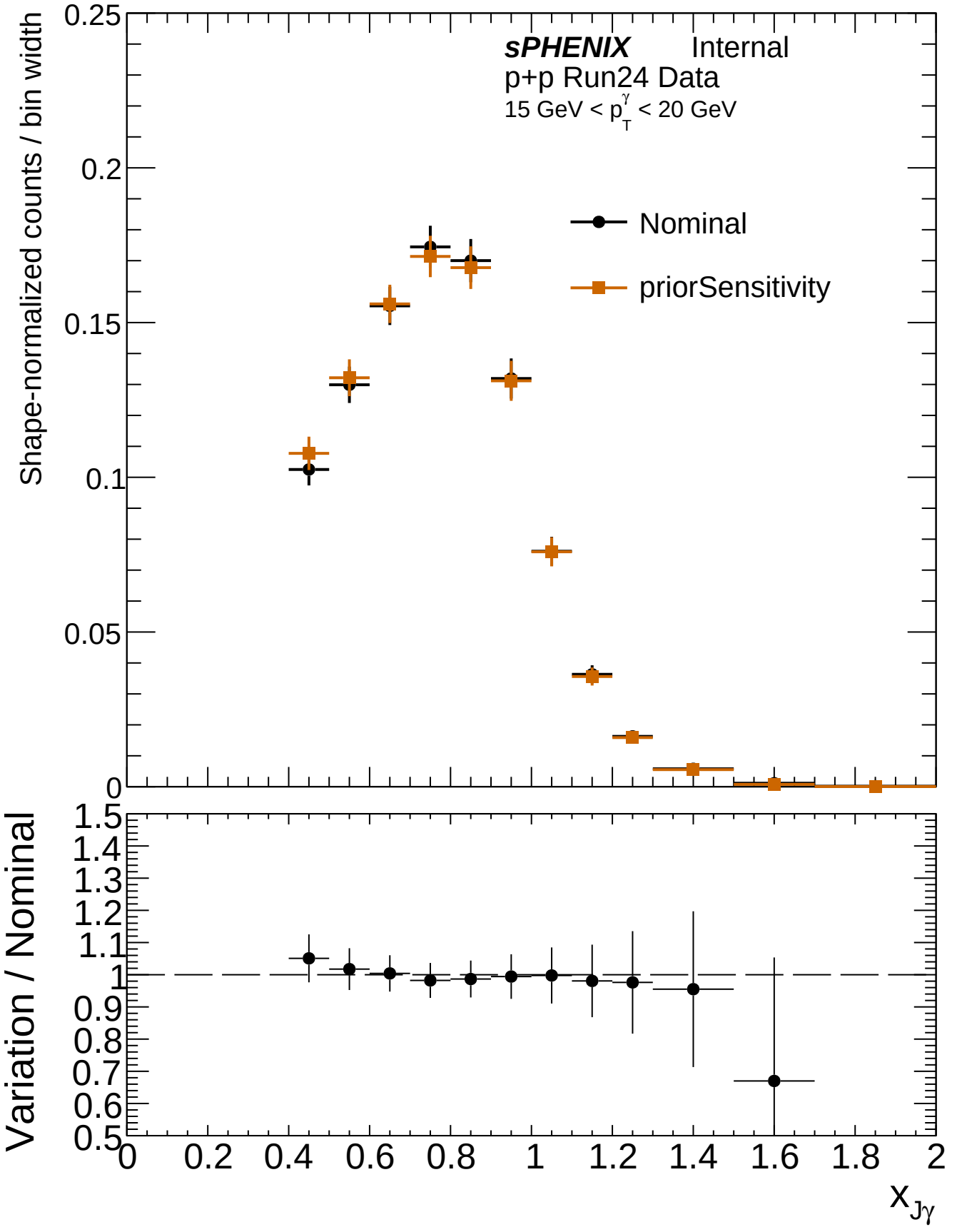

sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
niterLow
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

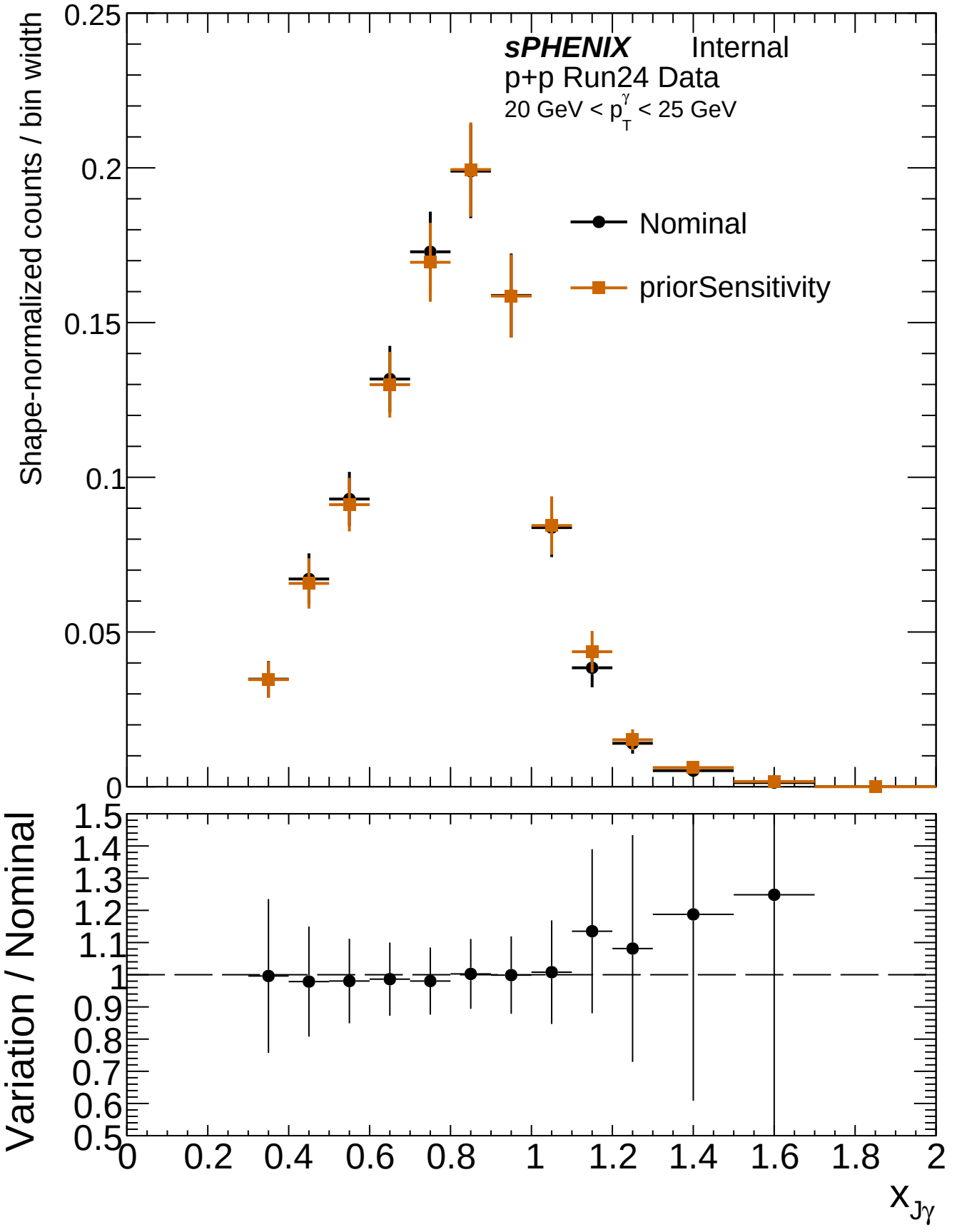

sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
niterLow
Shape-normalized counts / bin width
Variation / Nominal
$x_{J\gamma}$

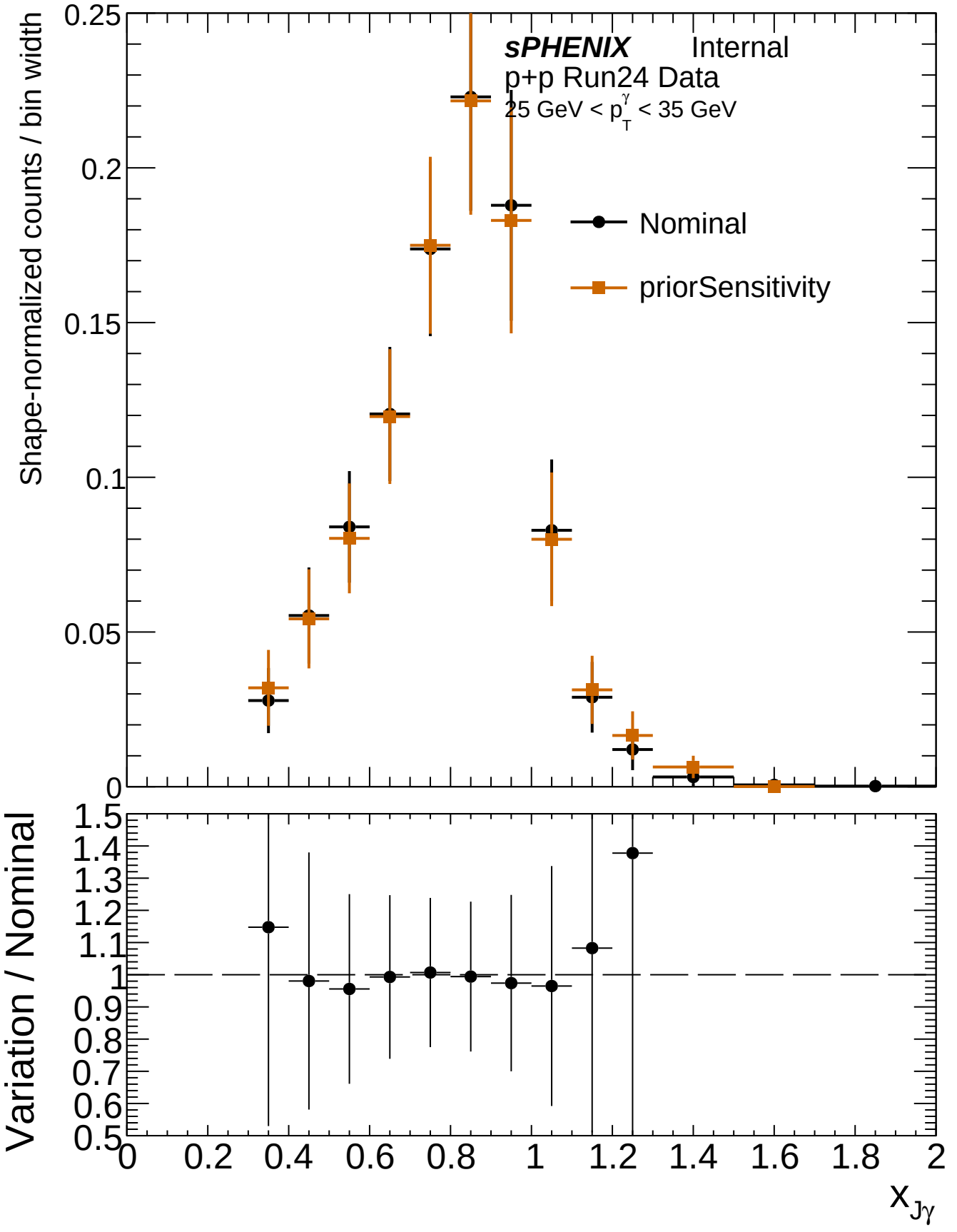

sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
niterHigh
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

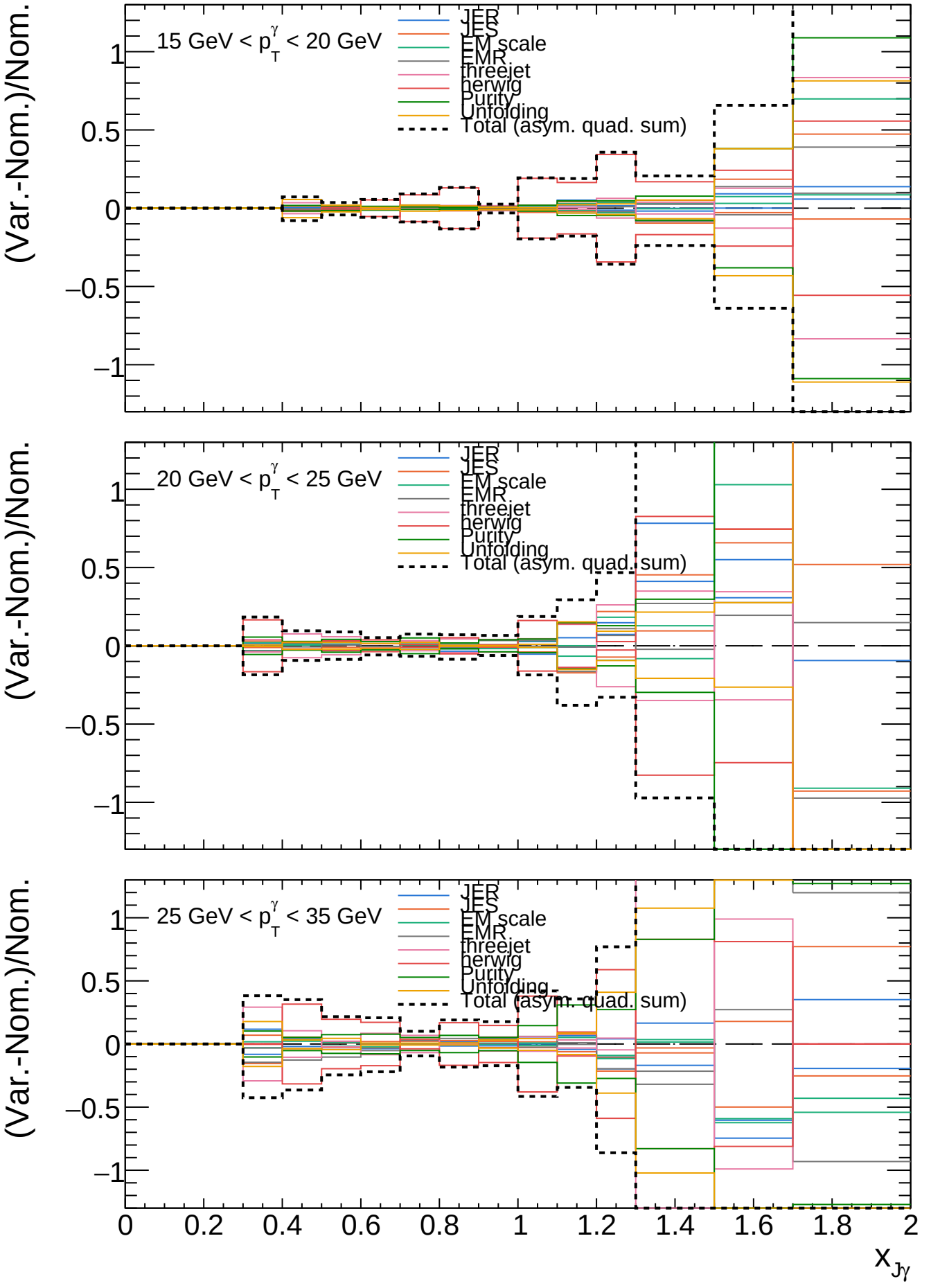

sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
niterHigh
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
niterHigh
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ
0.25
0.2
0.15
0.1
0.05
0
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2


sPHENIX Internal
p+p Run24 Data
15 GeV < p_T^γ < 20 GeV
Nominal
priorSensitivity
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ


sPHENIX Internal
p+p Run24 Data
20 GeV < p_T^γ < 25 GeV
Nominal
priorSensitivity
Shape-normalized counts / bin width
Variation / Nominal
x_Jγ


sPHENIX Internal
p+p Run24 Data
25 GeV < p_T^γ < 35 GeV
Nominal
priorSensitivity
Shape-normalized counts / bin width
Variation / Nominal
$x_{J\gamma}$


15 GeV < p_T^γ < 20 GeV
20 GeV < p_T^γ < 25 GeV
25 GeV < p_T^γ < 35 GeV
JER
JES
EM scale
EMR
threejet
herwig
Purity
Unfolding
Total (asym. quad. sum)
(Var.-Nom.)/Nom.
$x_{J\gamma}$
1
0.5
0
−0.5
−1
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2